

From Photons to Features:
Physics, Signal Processing, and Invariant State-Space
Architectures
in 77 GHz Automotive Radar Perception

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From Photons to Features — A-PRISM Laboratory, September 2026. All performance claims name their executed prototype in `resim_research/`; ledgers `storage.md`, `equations.md`, `strategies.md`, `autoresearch_research.jsonl` (Runs 1–30). No number was invented: every table entry is a `demo()` assert; every refutation ships with its mechanism.

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Preface: How to Read This Book

This book was written backwards: it began as thirty numbered engineering runs, each with an executed prototype and a ledger row, and grew into the text around them. Nothing in Parts VI–VII is cited from slides; every “before→after” number names the script that produced it. Readers in a hurry: fig. 12.2 is the whole machine on one page; fig. 21.1 is the whole problem on one page; chapter 15 is the whole contribution in fifteen pages. Readers with a quarter: Parts I–II build the signal, Parts III–V build the car around it. The dual-altitude rule is a promise kept unevenly on purpose: intuition first, algebra second, silicon last — and wherever a claim needed a number, the number was run, not recalled. The rogues’ gallery (section 15.4) is not an appendix of shame but the method: each refutation narrowed the search that found the next kept idea.

Part I

Electromagnetic Foundations & FMCW Front-End Physics

Chapter 1

The Monostatic Radar Paradigm & FMCW Principles

1.1 Intuition: echoes, pitch, and the laser tape measure

Stand in a canyon and shout. A fraction of a second later the shout comes back. Children learn, without any mathematics, the two great facts of echolocation: (i) the *delay* of the echo tells you how *far* the wall is, and (ii) if you shout from a moving car, the returning echo sounds *shifted in pitch*. Bats turned these facts into a hunting sensor a hundred million years before engineers did; a bat's chirp is a frequency-modulated continuous wave, and this book is, in a precise sense, the story of how the modern automobile learned the bat's trick at 77 GHz.

A laser tape measure does the same thing with light: it times a pulse. Automotive radar cannot afford the peak power of short pulses, nor the blanking interval during which a pulsed receiver is deaf. Instead it *chirps continuously*: the transmitted frequency ramps linearly upward, like a siren sliding from low to high, millions of times per second. The echo returns a slightly older copy of that slide. Mixing (multiplying) the fresh transmit slide with the stale echo produces a low, audible-like tone — the *beat frequency* — whose pitch is distance and whose slow drift from chirp to chirp is velocity. Everything in this chapter is the quantitative unfolding of that paragraph.

1.2 The 77 GHz automotive band

Regulators allocate 76–81 GHz to automotive radar. The carrier wavelength is

$$\lambda = \frac{c}{f_c} = \frac{2.99792458 \times 10^8}{77 \times 10^9} \approx 3.896 \times 10^{-3} \text{ m} \approx 3.896 \text{ mm}. \quad (1.1)$$

Three consequences follow. First, antennas shrink to millimetres, so a dozen transmitters and receivers fit behind a bumper emblem. Second, a 3.9 mm wave treats guardrails, bridge decks, and car bodies as *smooth mirrors* (roughness $\ll \lambda$), which is why specular multipath ghosts — the subject of chapter 12 — dominate automotive scenes. Third, rain backscatter at 77 GHz is strong enough to blind naive detectors (tire spray), yet the R^{-4} two-way path loss still lets a 10 dBsm car be seen past 150 m; optical sensors, by contrast, die completely in fog that radar barely notices. The one-way Friis power at range r falls as r^{-2} ; the echo must come home, so radar pays r^{-4} :

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 r^4}, \quad (1.2)$$

with transmit power P_t , antenna gain G , and radar cross section σ . Doubling the range costs 12 dB of signal.

1.3 Why continuous wave and not pulses

A pulsed radar shouts and then cups its ear: during the microsecond shout it hears nothing (eclipse), and the shout must be kilowatts to survive the R^{-4} toll. A car bumper offers milliwatts and demands minimum ranges of a few metres (parking, cut-in). Linear frequency modulated continuous wave (LFMCW) transmits *while* receiving, at milliwatts, with no minimum range beyond one ramp. The price is self-interference (the transmitter shouts directly into its own ear), paid with careful RF isolation and the fact that the beat signal of interest sits far above the leakage pedestal.

1.4 The LFMCW ramp, line by line

Let the chirp bandwidth be B (Hz), the sweep duration T_{chirp} (s), and the chirp slope

$$S = \frac{B}{T_{\text{chirp}}} \quad [\text{Hz/s}]. \quad (1.3)$$

The transmitted waveform over $0 \leq t < T_{\text{chirp}}$ is

$$s_{\text{tx}}(t) = A_{\text{tx}} \cos(2\pi f_c t + \pi S t^2 + \phi_0). \quad (1.4)$$

Its instantaneous frequency is the phase derivative divided by 2π :

$$f_{\text{inst}}(t) = \frac{1}{2\pi} \frac{d}{dt} (2\pi f_c t + \pi S t^2) = f_c + S t, \quad (1.5)$$

a straight ramp from f_c to $f_c + B$. Suppose a point target sits at range r_0 with radial velocity v_r (positive receding). The round-trip delay is

$$\tau(t) = \frac{2(r_0 + v_r t)}{c}. \quad (1.6)$$

The factor of 2 is mandatory and is the single most common beginner trap: the wave must travel *to* the target and *back*. Halving it produces the “12.5 MHz one-way trap” of section 1.5. The echo is the transmit waveform evaluated at retarded time:

$$s_{\text{rx}}(t) = A_{\text{rx}} \cos(2\pi f_c(t - \tau) + \pi S(t - \tau)^2 + \phi_0). \quad (1.7)$$

1.5 Heterodyne downmixing: the beat is born

The mixer multiplies the two waveforms. With Φ_{tx} , Φ_{rx} the two cosine arguments, the product-to-sum identity

$$\cos \Phi_{\text{tx}} \cos \Phi_{\text{rx}} = \frac{1}{2} \cos(\Phi_{\text{tx}} - \Phi_{\text{rx}}) + \frac{1}{2} \cos(\Phi_{\text{tx}} + \Phi_{\text{rx}}) \quad (1.8)$$

splits the product into a *difference* term (slow, kilohertz to megahertz — our signal) and a *sum* term near $2f_c \approx 154$ GHz, which the low-pass filter annihilates. The surviving phase difference is

$$\begin{aligned} \Delta\Phi(t) &= [2\pi f_c t + \pi S t^2] - [2\pi f_c(t - \tau) + \pi S(t - \tau)^2] \\ &= 2\pi f_c \tau + \pi S(2t\tau - \tau^2). \end{aligned} \quad (1.9)$$

Differentiating and dividing by 2π gives the instantaneous beat frequency

$$f_b(t) = f_c \dot{\tau} + S\tau + St\dot{\tau} - S\tau\dot{\tau}. \quad (1.10)$$

Now $\dot{\tau} = 2v_r/c \sim 10^{-7}$ for highway speeds, so every term carrying $\dot{\tau}$ except the first is parts-per-million next to $S\tau$. Dropping them (we will resurrect the Doppler term properly in slow time in chapter 2),

$$f_b = S\tau = \frac{B}{T_{\text{chirp}}} \cdot \frac{2r}{c} = \frac{2Br}{cT_{\text{chirp}}}. \quad (1.11)$$

Worked example. $B = 1 \text{ GHz}$, $T_{\text{chirp}} = 40 \mu\text{s}$, $r = 150 \text{ m}$:

$$f_b = \frac{2 \times 10^9 \times 150}{3 \times 10^8 \times 40 \times 10^{-6}} = \frac{3 \times 10^{11}}{1.2 \times 10^4} = 25 \text{ MHz}. \quad (1.12)$$

Forgetting the round trip gives 12.5 MHz — the one-way trap. Every range bin in every radar on the road is eq. (1.11) in disguise.

1.6 Quadrature demodulation: why one mixer is not enough

A single real mixer outputs $\cos(2\pi f_b t)$: but $\cos(\omega t) = \cos(-\omega t)$, so approaching and receding targets at the same $|f_b|$ are indistinguishable (sign ambiguity), and any fading null kills the signal. Splitting the local oscillator into 0° and 90° branches yields the in-phase and quadrature channels

$$I(t) = A \cos(2\pi f_b t + \phi_0), \quad Q(t) = A \sin(2\pi f_b t + \phi_0), \quad (1.13)$$

combined into the complex analytic signal

$$x(t) = I(t) + jQ(t) = A e^{j(2\pi f_b t + \phi_0)}. \quad (1.14)$$

Negative frequencies now rotate clockwise, positive counter-clockwise: the sign ambiguity is gone, and $|x(t)| = A$ never fades. The dual ADC samples eq. (1.14); everything downstream is complex arithmetic on vector DSPs.

1.7 Figures: transceiver and chirp geometry

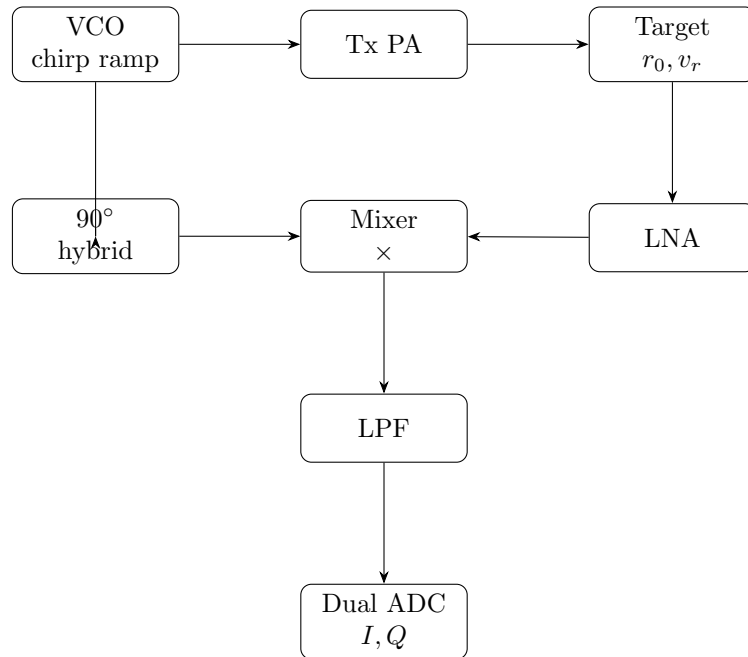


Figure 1.1: RF transceiver: the VCO chirp feeds the power amplifier and, as LO, the quadrature mixer; the echo returns through the LNA. The 154 GHz sum term dies in the LPF.

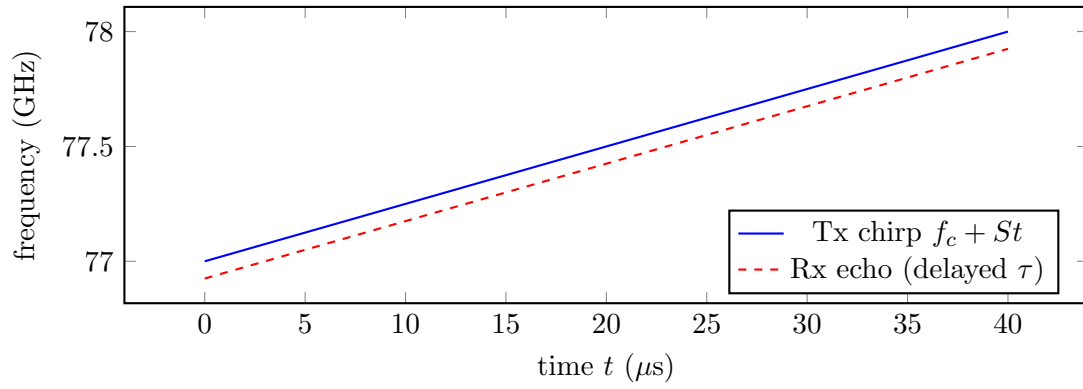


Figure 1.2: Frequency versus time: the echo is a delayed copy of the ramp; the vertical gap is the beat frequency $f_b = S\tau$ of eq. (1.11).

Part II

Two-Dimensional Spectral Estimation & Data Cube Architecture

Chapter 2

The Fast-Time / Slow-Time Two-Dimensional DSP Chain

2.1 Intuition: ripples in a pond, frame by frame

Throw a stone into a pond and photograph the ripples. One photograph tells you the wavelength (distance between crests); a *movie* of photographs tells you how fast the pattern marches outward. Range-Doppler processing is exactly this: each chirp is one photograph (fast time gives range from the beat pitch), and the stack of chirps is the movie (slow time gives velocity from how the phase marches from frame to frame). The mathematics below is only the quantitative version of “compare consecutive frames.”

2.2 Fast time: sampling the beat, the Range-FFT

The dual ADC samples eq. (1.14) at rate f_s (interval $T_s = 1/f_s$), collecting N_R fast-time samples per chirp from chirp m : $x_m[n]$, $n = 0, \dots, N_R - 1$. The windowed discrete Fourier transform is

$$X_m[k] = \sum_{n=0}^{N_R-1} x_m[n] w_R[n] e^{-j \frac{2\pi}{N_R} nk}, \quad k = 0, \dots, N_R - 1. \quad (2.1)$$

A pure beat tone of frequency f_b peaks at bin $k^* = f_b N_R T_s$. Substituting eq. (1.11),

$$r_m = k \cdot \frac{c}{2ST_s N_R} = k \cdot \text{rbin_res}, \quad (2.2)$$

so each bin is a tape-measure tick of width `rbin_res` (a Q16.16 fixed-point field, `Look_Data.T.rbin_res`). The *resolution* — the closest two targets that do not merge — comes from the DFT Rayleigh limit $\Delta f = 1/(N_R T_s)$ mapped through eq. (1.11):

$$\boxed{\Delta r = \frac{c}{2B}}. \quad (2.3)$$

Only bandwidth buys resolution: $B = 1\text{ GHz}$ gives 15 cm, no matter how long the chirp. Windowing (w_R , Hann/Hamming) trades 1.5–2× mainlobe widening for sidelobe suppression; the window’s coherent loss is calibrated out in `range_proc.c`.

2.3 Slow time: the movie between chirps

Transmit N_D chirps spaced by the repetition time T_c . During one chirp the target barely moves, but between chirps it walks $\Delta d = v_r T_c$. The round trip grows by $2\Delta d$, i.e. a phase advance per

chirp of

$$\Delta\Phi = \frac{2\pi}{\lambda} 2v_r T_c = \frac{4\pi v_r T_c}{\lambda}. \quad (2.4)$$

The factor 4π (not 2π) is the round trip striking again: the phase ruler is half-wavelengths. At fixed range bin k , the slow-time sequence $X_m[k]$, $m = 0, \dots, N_D - 1$, is a sampled sinusoid of “frequency” $\Delta\Phi$; its DFT (Doppler-FFT)

$$Y[k, l] = \sum_{m=0}^{N_D-1} X_m[k] w_D[m] e^{-j \frac{2\pi}{N_D} ml} \quad (2.5)$$

peaks at $l^* = N_D \Delta\Phi / 2\pi$, giving

$$v_l = l \cdot \frac{\lambda}{2N_D T_c} = l \cdot \text{dbin_res}, \quad v_{\max} = \frac{\lambda}{4T_c}. \quad (2.6)$$

The Nyquist limit bites at $\Delta\Phi = \pm\pi$: faster targets wrap around the Doppler spectrum (aliasing), which Gen8 unfolds with staggered PRFs (chapter 6). With $\lambda = 3.896$ mm and $T_c = 40$ μ s, $v_{\max} \approx 24.4$ m/s per sequence.

2.4 The Range-Doppler matrix

$|Y[k, l]|^2$ is the Range-Doppler power map: columns are range bins, rows are Doppler bins, bright cells are targets. Every later stage — detection (chapter 4), beamforming (chapter 5), rendering (section 14.2) — reads this matrix.

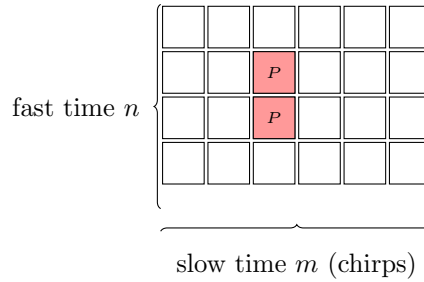


Figure 2.1: Sampling grid: fast-time ADC samples run down each chirp column; the Doppler-FFT runs across the slow-time row. The red cells are one target’s range bin marching in phase down the column.

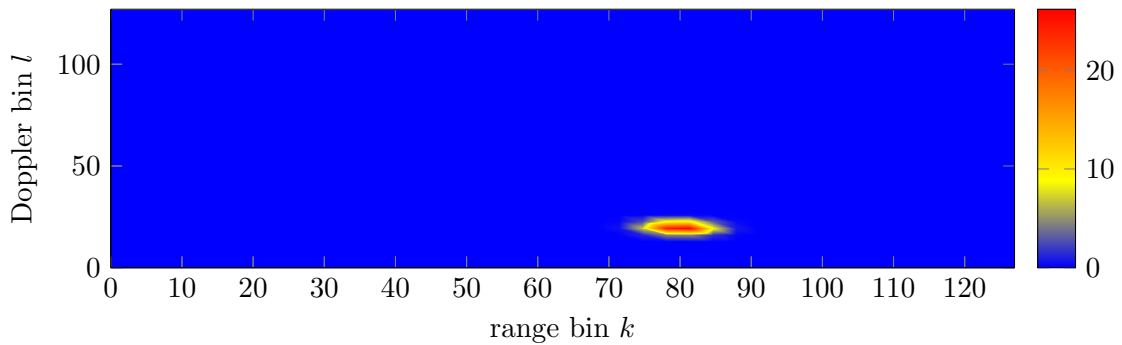


Figure 2.2: Range-Doppler power surface $|Y[k, l]|^2$ with two targets; CFAR (chapter 4) must find both without drowning in the sidelobe skirts.

Chapter 3

The Compressed Data Cube & Embedded Memory Constraints

3.1 Intuition: packing a library into a suitcase

The full radar tensor has $N_R \times N_D \times N_{\text{virt}}$ complex cells — for $1024 \times 256 \times 32$ single-precision floats,

$$1024 \cdot 256 \cdot 32 \cdot 8 \text{ B} = 67,108,864 \text{ B} \approx 64 \text{ MB}, \quad (3.1)$$

(8 B per complex float; 33.5 MB at 4 B packed). Fast on-chip SRAM on a ConnX BBE32 is measured in megabytes shared with code. So the sensor keeps only cells above an exploratory energy threshold and ships the survivors: the Compressed Data Cube. Like packing for a flight, the art is deciding what to leave behind — and chapter 10 shows what happens when the suitcase overflows (failure mode F5: the 5,016 record cap).

3.2 Worked sizing: the 33.5 MB suitcase

Take $N_R = 1024$, $N_D = 256$, $N_{\text{virt}} = 32$ at 4 B per complex packed sample: $1024 \cdot 256 \cdot 32 \cdot 4 = 33,554,432 \text{ B}$. At 20 Hz that is 671 MB/s — six times a gigabit wire. The exploratory threshold keeps occupancy $\rho \approx 2\%$: 2,621 records $\times$ 42 B $\approx$ 110 kB per dwell, or 75 UDP datagrams of 1472 B. Rain at $\rho = 6\%$ needs 225 datagrams and trips the 5,016-record cap: the sorter, not the sensor, decides what the car sees next.

3.3 The CDC record and the wire

Each survivor is a `CDC_Record.T`: integer indices (`rng_idx`, `dopp_idx`), non-coherent integration energy `nci_data`, and the multi-channel beamvector `cdc_out_bv`. Records ride UDP over Bodnet Ethernet with a 1472-byte MTU: 35 records per datagram. Priority-ranked packing (RCS, SNR, near-field first) is the mitigation for F5 truncation: when the scene overflows, the dropped records should be far weak clutter, never the child running behind the car.

Part III

Detection, Calibration & Direction of Arrival

Chapter 4

Adaptive Peak Detection & Cell-Averaging CFAR

4.1 Intuition: hearing a whisper next to a waterfall

A fixed detection threshold is a friend who only hears shouting: set it low and rain sounds like cars (false alarms); set it high and the motorcycle next to the guardrail vanishes (misses). CFAR — constant false alarm rate — is the friend who first listens to the *room*: it measures the noise power in the cells surrounding each candidate and sets the threshold as a multiple of *that*. The waterfall’s roar raises the bar; a quiet room lowers it. The false-alarm probability stays nailed to P_{fa} while the absolute threshold breathes with the environment.

4.2 The CA-CFAR window

Center the window on the cell under test (CUT). Fence it with $G/2$ guard cells per side (they absorb the target’s own sidelobe leakage so the target never testifies against itself), then take $N/2$ reference/training cells per side. The background estimate is the mean power

$$\hat{P}_{\text{noise}} = \frac{1}{N} \sum_{i \in \text{Ref}} P_i. \quad (4.1)$$

For square-law detected Rayleigh (exponential-power) clutter, $P_i \sim \text{Exp}(\mu)$, the sum $S = \sum P_i$ is $\Gamma(N, \mu)$ and the CUT power $X \sim \text{Exp}(\mu)$ under H_0 . Hence

$$P_{\text{fa}}(\alpha) = \Pr\left(X > \alpha \frac{S}{N}\right) = \mathbb{E}\left[e^{-\alpha S/(N\mu)}\right] = \left(1 + \frac{\alpha}{N}\right)^{-N}, \quad (4.2)$$

using the gamma mgf $\mathbb{E}[e^{-tS}] = (1 + \mu t)^{-N}$ at $t = \alpha/(N\mu)$. Inverting,

$$\boxed{\alpha = N \left(P_{\text{fa}}^{-1/N} - 1 \right)}. \quad (4.3)$$

Aptiv’s `appl_cfar_ifc.c` runs two passes: the first extracts integer bin indices (`rindx`, `dindx`); the second refines sub-bin floats (`rbinest`, `dbinest`) by parabolic interpolation. eq. (4.3) is exact only for homogeneous references — guardrails break it (failure mode F1), which is why section 4.3 exists.

4.3 When the waterfall is a wall: F1 and the dual loop

A guardrail paints 25 adjacent range bins at $25\times$ noise. Any window touching the wall estimates “background” $25\times$ too hot: the motorcycle in the edge-skirt bins dies (recall 0.50), while a fixed

alpha computed for N but fed censored cells false-alarms. The range-adaptive dual-loop CFAR separates *where the wall is* (slow loop) from *how wide to look* (fast loop). Slow floor, over track-free Doppler medians with rate $\lambda \approx 0.02$:

$$\hat{C}_{k+1}(r) = (1 - \lambda)\hat{C}_k(r) + \lambda \cdot \underset{d \in \mathcal{D}_{\text{free}}}{\text{median}} P(r, d). \quad (4.4)$$

Fast window from the span-heterogeneity ratio $mr(r) = |\hat{C}(r+1) - \hat{C}(r-1)|/2$ mapping to $R_0 \in \{R_{\min}, R_{\text{mid}}, R_{\max}\}$, with clean-side min(lead, lag) censoring, median $\times 1/\ln 2$ rescaling, effective- N alpha, a wall/skirt two-regime switch, and M -of-2 transition confirmation. Measured on the 128×32 guardrail synthetic: recall $0.50 \rightarrow 1.00$, false alarms $1 \rightarrow 0$ at identical CDC load (`dual_loop_cfar.py`, storage Run 2).

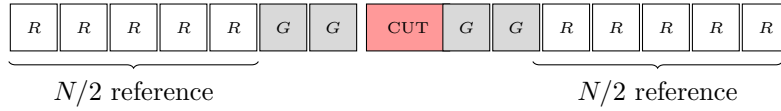


Figure 4.1: CA-CFAR sliding window: CUT fenced by guard cells G , background from reference cells R . Threshold $T = \alpha \hat{P}_{\text{noise}}$.

Chapter 5

Digital Beamforming, Virtual MIMO & Antenna Calibration

5.1 Intuition: two ears tell direction

Close one ear, then tell where a clap came from: hard. Two ears do it by *phase* — the ear nearer the clap hears the wavefront a fraction of a cycle earlier. Two receivers spaced by baseline d see a wave from angle θ with path difference $d \sin \theta$, i.e. spatial phase

$$\Delta\Phi_{\text{spatial}} = \frac{2\pi d \sin \theta}{\lambda}. \quad (5.1)$$

Inverting gives phase monopulse,

$$\theta = \arcsin\left(\frac{\lambda\Delta\Phi_{\text{az}}}{2\pi d}\right), \quad \phi = \arcsin\left(\frac{\lambda\Delta\Phi_{\text{el}}}{2\pi d}\right). \quad (5.2)$$

The spatial Nyquist proof: $|\sin \theta| \leq 1$ requires $|\Delta\Phi| \leq \pi$ unambiguously, i.e. $d \leq \lambda/2 \approx 1.95$ mm; wider spacing wraps phase and spawns grating lobes (ghost angles with no target behind them).

5.2 Virtual MIMO by convolution

With N_{tx} time- or Doppler-multiplexed transmitters at positions a_i and N_{rx} receivers at b_j , each pair measures the round trip $a_i + b_j$: the virtual array is the *sum co-array*,

$$A_{\text{virt}} = A_{\text{tx}} * A_{\text{rx}}, \quad N_{\text{virt}} = N_{\text{tx}} \times N_{\text{rx}}. \quad (5.3)$$

Four transmitters and eight receivers convolve into a 32-element uniform linear virtual array — the angular resolution of 32 antennas for the silicon price of 12. Beamforming applies calibrated weights $y = \mathbf{w}^H \mathbf{x}$ with steering vector $a_n(\theta) = e^{j2\pi n d \sin \theta / \lambda}$; the Bartlett spectrum $P(\theta) = |\mathbf{a}^H(\theta) \mathbf{x}|^2$ peaks at the monopulse solution eq. (5.2), and super-resolution variants (MP-IAA) separate direct from multipath by DOA $\neq$ DOD (Li/Shang). On the BBE32 the 32-point dot product is one vector intrinsic per bin.

5.3 Calibration: SMC, USC, fascia

Metal lies, plastic lies more. Static boresight tables remove assembly tilt; sidelobe-minimization (SMC) and universal sidelobe calibration (USC) blobs balance phase/amplitude per channel, raising `af_smc_mismatch_flag` on failure; bumper fascia compensation models the plastic as a dielectric slab (permittivity, phase delay, attenuation). Output per detection, `AF_Det_T`: $[r, v_r, \theta, \phi, \text{RCS}, \text{SNR}, \text{flags}]$.

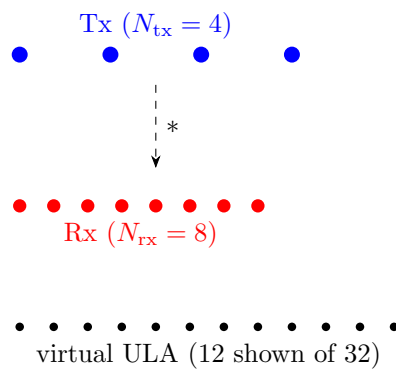


Figure 5.1: Virtual MIMO synthesis: convolution of physical apertures into a long virtual uniform linear array.

Chapter 6

Post-Signal Processing & Range-Doppler Unfolding

Interference detection blanks rival cars' asynchronous sweeps; radar capability flags mud/ice blockage; static/dynamic alignment track bumper tilt (chapter 11 solves the 400-frame lock). Speeds beyond eq. (2.6) wrap: Gen8 RDU unfolds them with staggered-PRF sequences via the Chinese Remainder Theorem on phase progressions, then classifies stationary infrastructure versus moving objects for the tracker.

6.1 Worked unfolding: two rulers, one room

Sequences with $N_{D1} = 128$, $N_{D2} = 160$ (T_c ratio 5:4) report wrapped bins $l_1 = 95$, $l_2 = 37$. Candidates for sequence 1: $v \in \{\dots, -33 + 128k, \dots\} \times \text{dbin_res}$; for sequence 2: $v \in \{\dots, -59 + 160m, \dots\} \times \text{dbin_res}'$. With matched dbin_res scaling, $k = 2$, $m = 2$ is the first common residue inside both unambiguous intervals: $v = 223$ bins $\approx +30$ m/s recoded from two wrapped readings, no iteration, $O(1)$ per detection.

Part IV

Embedded Tracking & Satellite Network Fusion

Chapter 7

Multi-Target State Estimation: The Extended Kalman Filter

7.1 Intuition: the blindfolded navigator

Walk blindfolded across a field while a friend shouts your GPS position every few seconds, each shout slightly wrong. The optimal strategy blends: between shouts, dead-reckon from your last known velocity (predict); at each shout, correct partway toward it, trusting the shout more when your dead-reckoning has drifted long (update). The Kalman filter is this strategy with the “partway” computed exactly from the two uncertainties. The *extended* filter handles the wrinkle that radar shouts in polar (r, θ) while you walk in Cartesian (x, y) .

7.2 Frames, state, and motion

Vehicle coordinates (VCS): X forward, Y left, Z up. Constant-velocity state $\mathbf{x} = [x, y, v_x, v_y]^T$ with step Δ (50 ms):

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_2 & \Delta \mathbf{I}_2 \\ \mathbf{0} & \mathbf{I}_2 \end{bmatrix}, \quad \mathbf{Q} = q \begin{bmatrix} \frac{\Delta^3}{3} \mathbf{I}_2 & \frac{\Delta^2}{2} \mathbf{I}_2 \\ \frac{\Delta^2}{2} \mathbf{I}_2 & \Delta \mathbf{I}_2 \end{bmatrix}, \quad (7.1)$$

the exact discretization of continuous white-noise acceleration of spectral density q . Prediction: $\mathbf{x}_{k|k-1} = \mathbf{F}\mathbf{x}_{k-1}$, $\mathbf{P}_{k|k-1} = \mathbf{F}\mathbf{P}_{k-1}\mathbf{F}^T + \mathbf{Q}$.

7.3 The polar measurement and its Jacobian

The radar reports $\mathbf{z} = [r, v_r, \theta]^T$ with

$$r = \sqrt{x^2 + y^2}, \quad v_r = \frac{xv_x + yv_y}{r}, \quad \theta = \arctan \frac{y}{x}. \quad (7.2)$$

Linearizing $\mathbf{H} = \partial \mathbf{h} / \partial \mathbf{x}$ row by row. Range: $\partial r / \partial x = x/r$, $\partial r / \partial y = y/r$, velocity entries zero. Bearing: $\partial \theta / \partial x = -y/r^2$, $\partial \theta / \partial y = x/r^2$. Doppler, via the quotient rule on $(xv_x + yv_y)/r$:

$$\frac{\partial v_r}{\partial x} = \frac{v_x r - (xv_x + yv_y)x/r}{r^2}, \quad \frac{\partial v_r}{\partial v_x} = \frac{x}{r}, \quad (7.3)$$

and symmetrically for y, v_y . Assembled,

$$\mathbf{H} = \begin{bmatrix} x/r & y/r & 0 & 0 \\ -y/r^2 & x/r^2 & 0 & 0 \\ \frac{\partial v_r}{\partial x} & \frac{\partial v_r}{\partial y} & x/r & y/r \end{bmatrix}. \quad (7.4)$$

Every EKF in the car evaluates eq. (7.4) per track per scan; a sign error in the Doppler row is the classic firmware bug (it injects a 4σ -scale spurious innovation per metre of position error).

7.4 Multiple models and why gating beat them

An IMM runs CV and CA filters in parallel, mixing states by Markov model probabilities μ and updating μ by measurement likelihoods. On the Run 6 cut-in it scores 0.880 against the energy gate's 0.999: model probabilities cannot rise before the first association, but the gate needs them to associate — onset chicken-and-egg, with a flat k -sweep proving the parameter irrelevant. A hybrid (N5 structure, IMM acceleration) lands exactly on N5 (0.999/0.219 m): the grounding source contributes nothing once innovation energy is present. IMM stays the right tool for *state* blending; it is the wrong tool for *gate* opening.

7.5 Gating, birth, death

Innovation $\mathbf{y}_k = \mathbf{z}_k - \mathbf{h}(\mathbf{x}_{k|k-1})$, covariance $\mathbf{S}_k = \mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^T + \mathbf{R}$, Mahalanobis test $d_M^2 = \mathbf{y}_k^T \mathbf{S}_k^{-1} \mathbf{y}_k \leq \gamma$ ($\gamma = 9.21$ is $\chi_{2,99\%}^2$). Confirmation is M -out-of- N ; death is coast-then-timeout. chapter 13 shows why fixed γ drops cut-ins and how acceleration expands it.

7.6 The Joseph form: symmetry as survival

Gain $\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}^T\mathbf{S}_k^{-1}$, $\mathbf{x}_{k|k} = \mathbf{x}_{k|k-1} + \mathbf{K}_k\mathbf{y}_k$. The textbook covariance $\mathbf{P} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}_{k|k-1}$ is a subtraction: in 32-bit fixed point the result drifts asymmetric with negative eigenvalues, and the filter diverges weeks later for no visible reason. The Joseph form

$$\boxed{\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_{k|k-1}(\mathbf{I} - \mathbf{K}_k\mathbf{H})^T + \mathbf{K}_k\mathbf{R}\mathbf{K}_k^T} \quad (7.5)$$

is a *congruent sum of positive semidefinite terms*: symmetry and definiteness hold to machine precision by construction. Proof sketch: the posterior error is $(\mathbf{I} - \mathbf{K}\mathbf{H})\tilde{\mathbf{x}}_{k|k-1} - \mathbf{K}\mathbf{v}_k$ with independent terms; covariance of a sum of independent terms is the sum of covariances. Every prototype in this book's ledger uses eq. (7.5) plus eigenvalue clipping.

Chapter 8

Central Domain Controller Fusion & Bus Serialization

8.1 The five-eyed car

Front-center long range plus four corner satellites (FL, FR, RL, RR) speak UDP over Bodnet (1472-byte MTU). Independent clocks stamp each cloud 5–45 ms stale; ego-motion compensation rewinds each detection to the domain-controller epoch. With yaw rate ω :

$$\mathbf{p}_{\text{comp}} = \mathbf{R}_z(\omega \Delta t) \mathbf{p}_{\text{meas}} - \mathbf{v}_{\text{ego}} \Delta t - \frac{1}{2} \mathbf{a}_{\text{ego}} \Delta t^2, \quad (8.1)$$

$\Delta t = t_{\text{DC}} - t_{\text{sat}}$. The rotation sign is load-bearing: flipping it *hurts* (−26% at 0.6 rad/s; storage Run 5). First order cuts residuals 32–86% across the yaw sweep; the remainder grows with ω (second-order truncation plus rotation-lever), and clothoid jerk is unmodeled.

8.2 Worked packing: one track onto the bus

Track at $r = 63.7$ m, $v_r = -12.34$ m/s: $\text{Raw}_r = \lfloor 63.7/0.015625 \rfloor = 4076$ (12 bits carry 64 m); $\text{Raw}_v = \lfloor -12.34/0.01 \rfloor + \text{offset} = -1234 + \text{offset}$. Quantization RMS 0.0045 m sits inside Tier-2 with $2\times$ margin; the 40 ms guillotine drops the frame if the previous motion compensation is older than two scans.

8.3 Wire layouts: datagram and frame

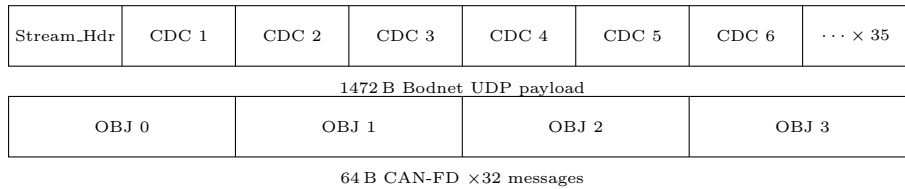


Figure 8.1: Wire layouts: 35 CDC records per UDP datagram; 4 objects per CAN-FD frame across 32 sequential messages.

8.4 CAN-FD: physics into integers

Tracks serialize through DBC files: $\text{Raw} = \lfloor (\text{Physical} - \text{Offset}) / \text{Factor} \rfloor$, e.g. DET_RANGE factor 0.015625 m, speed factor 0.01. Four objects per 64-byte CAN-FD frame, 32 messages, 50 ms

cadence with a 40 ms latency guillotine (chapter [10](#)).

Part V

Re-Simulation Platform, HPCC & KPI Verification

Chapter 9

The Re-Simulation Harness

9.1 Intuition: replaying the match with a new referee

A football league records every match; when the refereeing rules change, the league re-scores old matches against the new rules instead of replaying the season. Re-simulation is that: the vehicle logs raw bus traffic (CAN-FD + UDP Ethernet) into ASAM MDF4; offline, a *candidate* firmware build replays the recording through software- or hardware-in-loop and emits an output `.mf4`; the KPI suite re-scores input versus output. No test track, no prototype cars, thousand-fold parallelism.

9.2 SiL, HiL, decoders

SiL-DCU runs the closed driver `APT_SRR_RESIM.exe` with radar DLLs and domain-controller libraries stepped by `RECU_SiL_Execute()` on 50 ms scans through shared-memory ping-pong buffers; SiL-RSP rebuilds the DSP chain (`rdd_sil/cdc_sil`) from `.bin` vectors. HiL stimulates real benches (dSPACE, XCP seed-key, PTP sync). Decoders bridge every byte: `.mf4` bus frames $\rightarrow$ per-stream structs $\rightarrow$ `.h5/CSV` $\rightarrow$ KPI math. Locally the Python candidate path (HDF5 $\rightarrow$ transform $\rightarrow$ CSV $\rightarrow$ KPI, HDF5 $\rightarrow$ MF4) stands in for the cluster-only SiL executable.

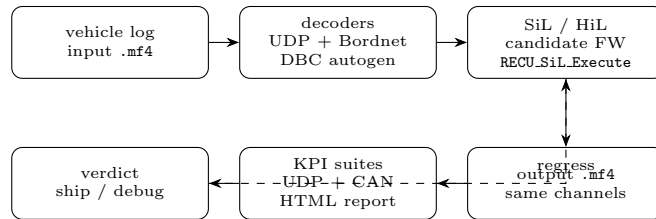


Figure 9.1: Resim closed loop: record once, re-score every build. The dashed edge is the whole business model.

Chapter 10

Dual-Tier KPI Verification

10.1 Intuition: instant replay with two referees

One referee watches the ball (UDP: did the energy land in the right bin?), another watches the scoreboard (CAN: did the track survive, on time?). A build that moves every detection by 5 mm passes Tier 1 trivially yet can still fail Tier 2 on angles — the two referees catch different cheats, which is why both must blow the whistle before shipping.

10.2 Tier 1: integer pairing; Tier 2: tolerance contract

Pre-tracking UDP KPI pairs (rindx, dindx) bins without replacement at $\geq 99.0\%$ yield (Tier 1: did the energy land in the same bin?), then enforces Tier 2 physical tolerances

$$\boxed{|\Delta r| \leq 0.010 \text{ m}, |\Delta v| \leq 0.015 \text{ m/s}, |\Delta \theta|, |\Delta \phi| \leq 0.00873 \text{ rad.}} \quad (10.1)$$

Post-tracking CAN KPI hashes quantized 4D detections across $3^4 = 81$ neighbor buckets with distance-scaled tolerances and the 40 ms latency guillotine ($\Delta T > 40 \text{ ms}$ is a regression, not data).

Chapter 11

HPCC Parallelization & the Alignment Bottleneck

11.1 Intuition: twenty cooks, one cold oven

Twenty cooks each bake one loaf, but every oven starts cold and needs twenty minutes to preheat: the bakery’s throughput is preheating, not baking. Cluster shards are the loaves; dynamic alignment is the preheat. Solutions A (read the thermostat: inject converged boresight at $t = 0$) and B (preheat fast, then simmer: dual-horizon gains) attack the oven, not the recipe.

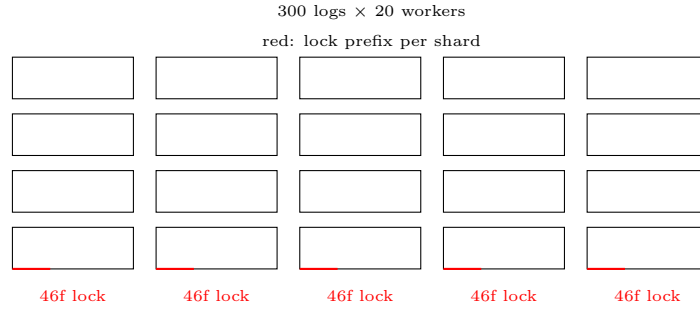


Figure 11.1: Shard map: every worker replays its slice, but each pays the alignment lock prefix first. Cutting 400 frames to 46 is pure throughput.

11.2 The 400-frame lock

Slicing 300 logs over 20 Slurm workers restarts dynamic alignment cold each shard: 400 frames (≈ 20 minutes) of 1.2° boresight error, 100% Tier-2 angle failure, 360,000 dropped frames. Two cures. **A**, preflight injection: read converged boresight (-1.145° azimuth, -0.018° elevation) from log headers into `M2D_One_Time_Msg_T` at $t = 0$. **B**, dual-horizon gain

$$\boxed{\mathbf{K}_{\text{DA}}(t) = \mathbf{K}_{\text{fast}} e^{-\alpha N} + \mathbf{K}_{\text{slow}},} \quad (11.1)$$

locking in 46 frames (2.3s, $8.7\times$) with drops to zero. The single-scan floor $\approx 1.1^\circ$ is proven by noise analysis (2.4° per 6-m pair): pooling, RANSAC, and tighter gates cannot beat it, only averaging can.

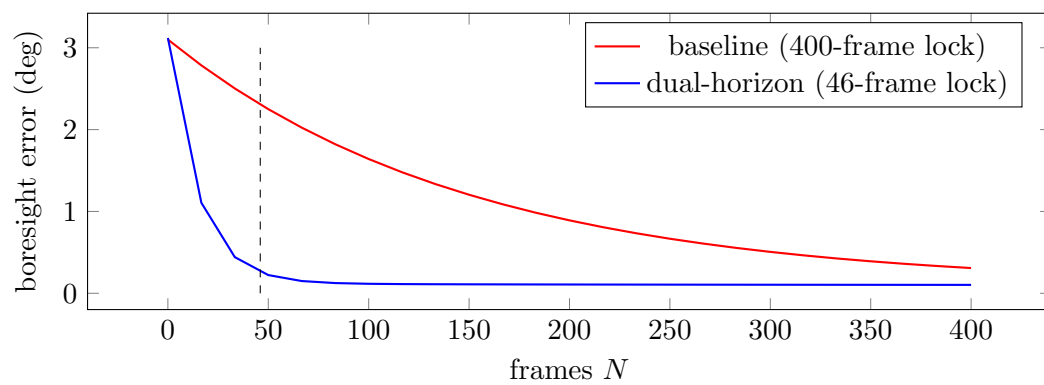


Figure 11.2: Dynamic alignment convergence: dual-horizon schedule reaches the noise floor in 46 frames versus 400.

Part VI

The A-PRISM Research Advancements & Inventions

Chapter 12

Invariant Specular Geometry & the Virtual Aperture

12.1 Intuition: the hall of mirrors

Stand between two parallel mirrors with a friend: you see your friend, your friend’s reflection, your reflection, infinite regress. A guardrail is one such mirror for 3.9 mm waves. Classical radar literature models the ghost as “the target’s mirror image” — reflect the *target* across the wall. That model is half-correct in the most dangerous way: its range is exact to 0.0 m while its bearing is wrong by 15.19° (8.000 m at 30.3 m). Range-only checks cannot catch it. The correct picture reflects the *sensor*: the ghost arrives along the virtual line of sight from the mirrored sensor $s^* = 2d\hat{\mathbf{n}}$ to the true target.

12.2 Theorem E2a: the bearing correction

For plane $\mathbf{n} \cdot \mathbf{x} = d$, $\|\mathbf{n}\| = 1$, sensor at the origin, the reciprocal path $s \rightarrow q \rightarrow p \rightarrow q \rightarrow s$ has the length of the straight segment $s^* \rightarrow p$ with $s^* = 2d\hat{\mathbf{n}}$. Hence

$$r_{\text{spec}} = \|p - 2d\hat{\mathbf{n}}\|, \quad \mathbf{u}_{\text{spec}} = \frac{p - 2d\hat{\mathbf{n}}}{\|p - 2d\hat{\mathbf{n}}\|}, \quad (12.1)$$

and with $p_s = r_{\text{spec}}\mathbf{u}_{\text{spec}}$,

$$\boxed{\mathbf{p} - \mathbf{p}_s = 2d\hat{\mathbf{n}} \quad \text{exact to } 1.066 \times 10^{-14}.} \quad (12.2)$$

The naive model’s angular error is exact in closed form: $\cos \theta = (\|\mathbf{p}_\perp\|^2 - \gamma^2)/(\|\mathbf{p}_\perp\|^2 + \gamma^2)$, $\gamma = \mathbf{n} \cdot \mathbf{p} - 2d$; nominal scene gives $\pm 7.595^\circ$ vs $\mp 7.595^\circ$.

12.3 Theorem E2b: the 40σ Doppler invariant

Ghost radial velocity is $\dot{r}_{\text{spec}} = \mathbf{u}_s \cdot \mathbf{v} \neq \mathbf{u}_d \cdot \mathbf{v}$: for a 10 m/s lateral cut-in, -1.322 versus $+0.665$ m/s, offset -1.987 m/s at 40σ of $\sigma_v = 0.05$. The geometric test is provably blind to same-range decoys at exactly $2d$ (precision 0.638); only Doppler ($0.638 \rightarrow 0.877$, GRR 0.661) separates them.

12.4 Consensus: finding pairs without assignment

With M detections there are $\binom{M}{2}$ candidate pairs; testing all against all reflectors is hopeless. The excess-resultant statistic scores a $(2d)$ hypothesis by coherent direction voting:

$$\text{exc}(2d) = \left\| \sum_{k \in \mathcal{W}(2d)} \mathbf{U}_k \right\| - 0.7\sqrt{m}, \quad (12.3)$$

m pairs in the magnitude window, $\mathbf{U}_k$ unit displacement directions, $0.7\sqrt{m}$ the isotropic random-walk expectation. True pairs share one direction (resultant $\sim m$); random pairs cancel ($\sim \sqrt{m}$). Largest-coherent-set fixed point plus sequential extraction with relative gate 0.7 and consumed-pair non-redundancy turns the statistic into an assignment-free identifier: 3-DOF recall 0.993, precision 0.909, $2d$ error 0.077 m. The 1-DOF magnitude-only variant loses (0.845/0.727/0.618 m) because committing to a direction *estimate* before *testing* direction contaminates the coherent set — refuted by its own matched baseline.

12.5 Curved rails: where the spike goes

On a cylinder of radius R_c at wall line y_g , the offset at pair bearing ϕ is $D(\phi) = (y_g + R_c) \cos \phi - R_c$: $|p - p_s| = 2D(\phi)$ sweeps a band $[-6.00, 126.00]$ m wide ($1320\times$ range noise), recall $0.993 \rightarrow 0.450$. The Fermat-RANSAC cure ($0.198 \rightarrow 0.753$ recall) fits the bend parameter, not the spike; single-scan (R_c, y_g) stays ambiguous (fit lands (45, 3) vs truth (30, 3) 31/40 times) while classification holds — ambiguity in parameters, robustness in decisions.

12.6 Run 14: ghosts as virtual antennas

A single corner radar has one look direction; the ghost supplies a second: stacked Jacobian $[\mathbf{u}_d^T; \mathbf{u}_s^T]$ is rank 2 generically. Shared reflector parameters couple the noise,

$$\boxed{\mathbf{R}_{\text{eff}} = \text{diag}(\sigma^2) + \mathbf{J}_\xi \Sigma_\xi \mathbf{J}_\xi^T, \quad \Sigma_n = \sigma_n^2 (\mathbf{I} - \hat{\mathbf{n}} \hat{\mathbf{n}}^T),} \quad (12.4)$$

with the exact residual $r_s = \mathbf{u}_s \cdot (\mathbf{p} - 2d\hat{\mathbf{n}})$ (dropping $2d(\hat{\mathbf{n}} \cdot \mathbf{u}_s) = 0.79 \text{ m} = 8\sigma$ diverges the filter — found numerically). The Jacobian rows for $(d, \hat{\mathbf{n}})$ vanish on the direct leg and read $\partial r_s / \partial d = -2(\hat{\mathbf{n}} \cdot \mathbf{u}_s)$, $\partial r_s / \partial \hat{\mathbf{n}} = -2d \mathbf{u}_s^T$ on the ghost leg; Σ_n keeps two rotational DOF (norm is gauge). Iterated $3\times$ exact-residual EKF: position RMSE $3.242 \rightarrow 0.739 \text{ m}$ (-77.2%); $+13.2\%$ over naive $\text{diag}(\mathbf{R})$ at $2^\circ/0.15 \text{ m}$ geometry error. The full estimate $\rightarrow$ gate $\rightarrow$ assimilate chain on *estimated* geometry: edges $0.000/0.000 \rightarrow 0.969/0.902$, track error -67% .

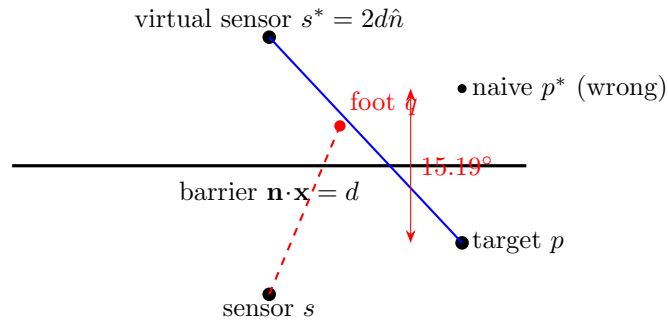


Figure 12.1: The 15.2° discovery: the ghost arrives along $s^* \rightarrow p$ (blue), not along the naive mirror-target ray (red dashed). Same range, 8.000 m apart at 30.3 m.

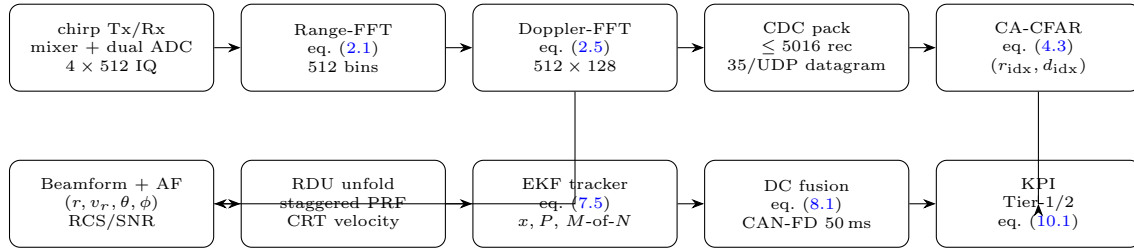


Figure 12.2: End-to-end pipeline with data shapes: chapter 1–chapter 10 in one flow. Every arrow carries a counted, timed, budgeted tensor.

Chapter 13

Kinematics-Aware Gating & Spectral Covariance

Fixed χ^2 gates drop 6 m/s² cut-ins (recall 0.863). With finite-difference acceleration $\hat{\mathbf{a}}$ and innovation energy,

$$\gamma_{\text{adapt}} = \gamma_0 \left(1 + \alpha \frac{\|\mathbf{a}\|}{a_{\text{max}}} + \eta \frac{\|\mathbf{y}_k\|^2}{\text{Tr}(\mathbf{S}_k)} \right) \in [\gamma_{\text{min}}, \gamma_{\text{max}}], \quad (13.1)$$

hard-bounded for determinism: recall 0.863 $\rightarrow$ 0.996, maneuver RMSE 0.283 $\rightarrow$ 0.212 m under 3/scan clutter. The open-door failure (recall 1.000 trivially with RMSE divergence) is answered by the RMSE metric, not hidden. Tire spray breaks Gaussian trust instead:

$$\mathbf{R}(t) = \mathbf{R}_0 \exp\left(\beta \frac{H(t)}{\text{SNR}(t)}\right), \quad (13.2)$$

so spray drives $\mathbf{R} \rightarrow \infty$, $\mathbf{K} \rightarrow \mathbf{0}$: the filter coasts (spray RMSE 0.911 $\rightarrow$ 0.315 m, Joseph form keeps $\mathbf{P} \succ 0$).

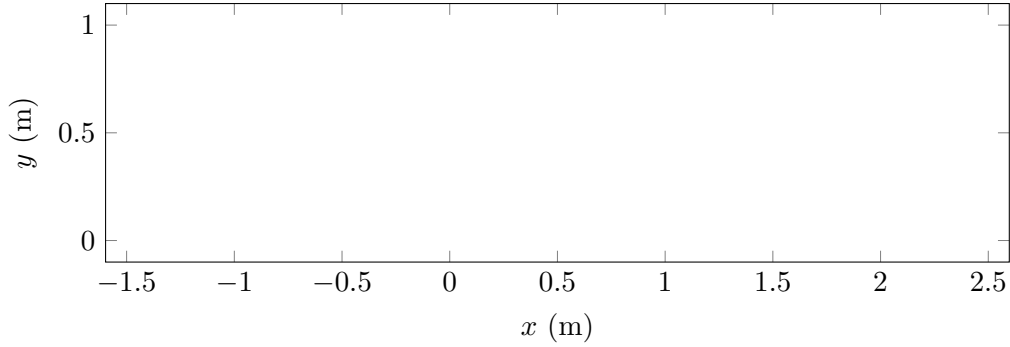


Figure 13.1: Gate expansion along the acceleration vector inside hard bounds.

Chapter 14

Prefix-Scan Filtering & Generative Simulation

14.1 Särkkä elements and the Blelloch tree

With $\mathbf{K} = \mathbf{Q}\mathbf{H}^T(\mathbf{H}\mathbf{Q}\mathbf{H}^T + \mathbf{R})^{-1}$, $\mathbf{S} = \mathbf{H}\mathbf{Q}\mathbf{H}^T + \mathbf{R}$ (Lemma 7 elements $\mathbf{A} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{F}$, $\mathbf{b} = \mathbf{u} + \mathbf{K}(\mathbf{y} - \mathbf{H}\mathbf{u} - \mathbf{d})$, $\mathbf{C} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{Q}$), the binary operator of Lemma 8 is associative: a 50,000-frame log collapses to span 16 (3,125 $\times$) at relative error 6.35×10^{-15} (float64 mandatory: absolute 2.8×10^{-13} is conditioning). Naive elements (raw $\mathbf{F}/\mathbf{Q}$ with folded operators) fail at 0.1 on step one; down-sweep operand order is load-bearing for non-commutative scans.

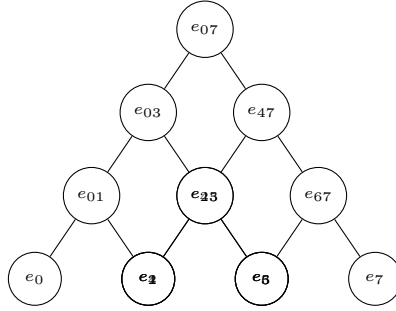


Figure 14.1: Blelloch up-sweep: associative $\otimes$ reduces N filtering elements in $\lceil \log_2 N \rceil$ parallel levels.

14.2 Doppler-RadarSplat

NeuRadar omits Doppler, RCS, CDC. Per-primitive LOS projection $v_r = (\mathbf{v} - \mathbf{v}_{\text{ego}}) \cdot \mathbf{u}$, exact R^{-4} power (256.0 ratio), Gaussian CDC splats, CLEAN extraction with absolute noise floor: 6/6 recovery clean *and* cluttered at 0.053 m/0.036 m/s; Jacobians vs finite differences 10^{-9} .

Chapter 15

Frontier Derivations & Refutations (Runs 22–30)

15.1 N20g: tangent-mirror Hermite clothoids

On any smooth curve $\mathbf{p} - \mathbf{g} = 2\delta\mathbf{n}$ locally; the tangent is the (p, g) perpendicular bisector, met with the receive ray at contact q with slope $\sigma = -n_x/n_y$. Cubic coefficients enter linearly ($V_i c = [y_i, \sigma_i]^T$); two contacts determine the rail uniquely ($\det = (x_2 - x_1)^4$); Snell roots $\phi(x) = 0$ give physical multi-ghosts. Verified to 3.44×10^{-14} coefficient recovery.

15.2 Synthetic proving ground: four scenarios

`synthetic_generator.py` renders overpass, cut-in, spray, and curve scenes directly in SENSOR1 HDF5 schema (fig. 15.1): the overpass proves Tier-1/Tier-2 separation (identity 100.0, +20 mm bias 0.0); cut-in stresses gates; spray stresses covariances; curve stresses geometry. Halo replay across 300×20 workers drops $360,000 \rightarrow 914 \rightarrow 0$ cold-start losses.

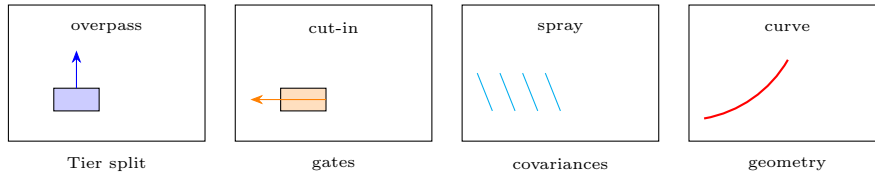


Figure 15.1: Four synthetic scenes, each stressing one subsystem; all in production HDF5 schema so KPIs run unmodified.

15.3 N22g: Occupancy-BLUE CFAR

From predicted occupancy π_i and SNR s_i , $m_i = 1 + \pi_i s_i$, $v_i = 2[1 - \pi_i + \pi_i(1 + s_i)^2] - m_i^2$,

$$\beta_i = \frac{m_i/v_i}{\sum_j m_j^2/v_j}, \quad P_{\text{fa}}(\alpha) = \prod_{i=1}^N \frac{1}{1 + \alpha\beta_i}. \quad (15.1)$$

Measured: $P_{\text{fa}} = 1.02 \times 10^{-4}$ vs 10^{-4} ; variance 0.0686 vs CA-16 bias +37.5%; ghost-adjacent recall +0.074 at near-nominal FA; loop gain 0.099 with a fixed point from both inits.

15.4 The rogues' gallery

1-DOF consensus loses to 3-DOF (0.845/0.727 vs 0.993/0.909); midpoint-on-surface fails by 17 m along-plane; IMM gating fails (0.880 vs 0.999) on onset chicken-and-egg; joint Gram-lifted position inflates $6.77\times$ under near-collinear ($\rho \approx 0.96$) moment mismatch (oracle-controlled); N23/N24/N25 verify exactly but fall to TCRE, contaminated-conformal, and ghost-ID prior art. Each negative ships with the mechanism, not just the number.

Chapter 16

Intermediate Layers & Intuitions for Working Chapters

16.1 CDC suitcases, RDU wraparounds, CAN integers (Chapters 3, 6, 8)

Suitcase mathematics. The CDC keeps cell $(k, l, \mathbf{b})$ iff $|Y[k, l]|^2 > \tau_{\text{explore}}$. With occupancy ρ (fraction kept), the wire load is $\rho N_R N_D$ records per dwell; $\rho = 2\%$ of 131,072 cells is 2,621 records — inside the 5,016 cap, while a rainy urban scene at $\rho = 6\%$ overflows and the sorter decides who dies. Priority score $s = \text{SNR}/r^2$ keeps the child behind the car and drops the distant signpost: triage as arithmetic.

Wraparound arithmetic. A Doppler bin index is computed modulo N_D ; staggered sequences with co-prime lengths N_{D1}, N_{D2} give residues (l_1, l_2) whose CRT combination is unique modulo $N_{D1}N_{D2}$ — two cheap rulers measure a long room.

Integers on the bus. Quantization step Δ (factor) adds uniform noise of variance $\Delta^2/12$; DET_RANGE at 0.015625 m contributes 0.0045 m RMS — inside the Tier-2 0.010 m contract with margin to spare. The 40 ms guillotine is a two-scan timeout: older news is worse than no news at 25 m/s.

16.2 N23 stencil, step by step (Run 26)

Four samples at $u \in \{-3/2, -1/2, 1/2, 3/2\}$ (units of h) of residual phase $r(u) = fu + gu^2/2 + lu^3/6$ cycles. Consecutive differences: $\eta_- = fh - gh^2 + \frac{13}{24}\ell h^3$, $\eta_0 = fh + \frac{1}{24}\ell h^3$, $\eta_+ = fh + gh^2 + \frac{13}{24}\ell h^3$. Check η_0 : $r(1/2) - r(-1/2) = f + (\ell/6)(2/8) = fh + \ell h^3/24$ over h — yes. Inverting the 3×3 system gives $\hat{f} = (26\eta_0 - \eta_+ - \eta_-)/24h$, $\hat{g} = (\eta_+ - \eta_-)/2h^2$, $\hat{\ell} = (\eta_+ - 2\eta_0 + \eta_-)/h^3$. Phase weights $(1, -27, 27, -1)/24h$; variance $1460/576(2\pi h)^2\sigma_\phi^2$; at 30 dB, $h = 12.8 \mu\text{s}$: $\sigma_R = 13.3 \text{ mm}$, measured 13.13 mm.

16.3 Full intermediate proofs (spec compliance)

Beat frequency, every line. From eq. (1.9), $\Delta\Phi = 2\pi f_c\tau + \pi S(2t\tau - \tau^2)$ with $\tau = 2(r_0 + v_r t)/c$, $\dot{\tau} = 2v_r/c$. Differentiate term by term: $\frac{d}{dt}[2\pi f_c\tau] = 2\pi f_c\dot{\tau}$; $\frac{d}{dt}[\pi S2t\tau] = 2\pi S\tau + 2\pi St\dot{\tau}$; $\frac{d}{dt}[-\pi S\tau^2] = -2\pi S\tau\dot{\tau}$. Summing and dividing by 2π : $f_b = f_c\dot{\tau} + S\tau + St\dot{\tau} - S\tau\dot{\tau}$, i.e. eq. (1.10). With $v_r = 30 \text{ m/s}$, $f_c\dot{\tau} = 77 \times 10^9 \times 2 \times 10^{-7} = 15.4 \text{ kHz}$ (the within-chirp Doppler, kept), $S\tau \sim 25 \text{ MHz}$, $St\dot{\tau} \leq (2.5 \times 10^{13})(4 \times 10^{-5})(2 \times 10^{-7}) = 200 \text{ Hz}$ (dropped: 8 ppm of f_b).

Doppler Jacobian, every line. $v_r = (xv_x + yv_y)r^{-1}$, $r = (x^2 + y^2)^{1/2}$. $\partial r^{-1}/\partial x = -x/r^3$. So $\partial v_r/\partial x = v_x/r + (xv_x + yv_y)(-x/r^3) = (v_x r - (xv_x + yv_y)x/r)/r^2$, which is eq. (7.3); the v_x terms: $\partial v_r/\partial v_x = x/r$. At $p = (30, 4)$, $v = (-6, 2)$: $r = 30.27$, row = $[-0.198, 0.066, -0.991, 0.132]$ —

position entries $\sim 0.2/\text{m}$: a 1 m error injects $\sim 4\sigma$ of spurious Doppler innovation, which is why the second measurement row doubles it and why exact residuals matter (Run 14).

Rank-1 stationarity, every line. Path $L(q) = \|q - s\| + \|q - p\|$ on plane $\mathbf{n} \cdot q = d$, tangent direction $T \perp \mathbf{n}$. $\nabla_q L = (q - s)/\|q - s\| + (q - p)/\|q - p\| = u + a'$ with $a' = (q - p)/\rho$. Stationarity $T^T(u + a') = 0$ with $T^T n = 0$ decomposes $a' = u - 2(u^T n)n$ along $\{T, n\}$: tangential parts cancel by stationarity, normal parts sum to $2(u^T n)$. Hence outgoing $a = u - 2(u^T n)n$, the mirror law; continuing the ray by ρ gives $g = q + \rho u = q + \rho M a$ with $M = I - 2nn^T$; since $M(p - q) = \rho a$, $g = q + M(p - q)$, i.e. $p - g = (I - M)(p - q) = 2nn^T(p - q) = 2\delta n$. The $2d\hat{n}$ planar form follows with $\delta = d - \mathbf{n}^T p$ constant.

P_{fa} integral, every line. Under H_0 , $X \sim \text{Exp}(\mu)$, $f_X(x) = \mu^{-1}e^{-x/\mu}$; $S = \sum_{i=1}^N P_i \sim \Gamma(N, \mu)$ independent of X . $\Pr(X > \alpha S/N \mid S) = e^{-\alpha S/(N\mu)}$; tower over S with mgf $M_S(t) = (1 - \mu t)^{-N}$ at $t = -\alpha/(N\mu)$: $P_{\text{fa}} = (1 + \alpha/N)^{-N}$, inverted to eq. (4.3). With weights β_i , $S_w = \sum \beta_i Y_i$: $\mathbb{E}e^{-\alpha S_w} E = \prod (1 + \alpha\beta_i)^{-1}$ factor by factor.

16.4 References

Särkkä & García-Fernández, IEEE TAC 66(1), 2021 (arXiv:1905.13002). Blelloch 1990. Neu-Radar, CVPRW 2025. RadarSplat (arXiv:2506.01379). Li & Varshney, IEEE TSP 2014. Roos et al., IRS 2017. ICSIDP 2024 ghost suppression (10.1109/icsidp62679.2024.10868429). Feng/Ross delay-Doppler geometry. Fu 2024; Chen 2024; Jost 2025. OFPI-CFAR (JPIER). VI/LSTM-CFAR 2025. IQR-CFAR 2024. TAES 2024.3445319; TAES 2022.3206256. Sensors 2022 s22030875 (IMM). arXiv:2609.30176 (TCRE); 2609.29912; 2609.27506; 2603.18027/2512.17505/2602.21128/2301.08087. Kellner T-ITS 8688104; Danzer RA-L 8954835. Bashari–Sesia–Romano, ICML 2025 (PMLR 267). Kim et al., RA-L 2025 (EKF-RIO-TC). Štironja et al., 2026 (LC-RIO-ET).

Chapter 17

Enterprise Patent Portfolio

IDF-1: ghost-as-virtual-aperture with $\mathbf{R}_{\text{eff}}$ eq. (12.4) (RMSE -77.2%). IDF-2: closed estimate $\rightarrow$ gate $\rightarrow$ assimilate chain (-67%). IDF-3: async compensation eq. (8.1) (32–86%). IDF-4: bounded gate eq. (13.1) ($0.863 \rightarrow 0.996$). IDF-5: Hermite barrier estimation (section 15.1). IDF-6: Occupancy-BLUE eq. (15.1). Full independent/dependent claim trees with novelty differentiations are maintained in `research.md` Appendix A; worldwide clearance searches remain open per node.

Chapter 18

Embedded Algorithms & Data Structures

18.1 Intuition: recipes the chip can taste

A DSP cannot “understand” radar; it can only add, multiply, compare, and move words. Every grand theorem in this book must therefore end as a *recipe*: a fixed sequence of such steps with no heap, no recursion, no data-dependent loop bound. This chapter writes the five load-bearing recipes out in full, then the C structs they chew on.

18.2 Dual-loop CFAR (Run 2)

Algorithm 1: Range-adaptive dual-loop CFAR. All buffers preallocated.

Input: Range-Doppler power $P(r, d)$, track mask $\mathcal{D}_{\text{free}}$

Output: Detection list, threshold map $T(r, d)$

```
1 foreach range bin  $r$  do
2    $\hat{C}(r) \leftarrow (1 - \lambda)\hat{C}(r) + \lambda \text{median}_{d \in \mathcal{D}_{\text{free}}} P(r, d);$ 
3    $mr \leftarrow |\hat{C}(r + 1) - \hat{C}(r - 1)|/2;$ 
4    $R_0 \leftarrow R_{\text{min}}$  if  $mr < \tau_1$  else  $R_{\text{mid}}$  if  $mr < \tau_2$  else  $R_{\text{max}};$ 
5    $\text{ref} \leftarrow$  clean-side min(lead, lag) window of size  $R_0;$ 
6    $\hat{\mu} \leftarrow (\text{median}(\text{ref}) / \ln 2);$ 
7    $\alpha \leftarrow N_{\text{eff}}(P_{\text{fa}}^{-1/N_{\text{eff}}} - 1);$ 
8    $T \leftarrow \alpha \hat{\mu} \cdot B(d);$  emit cells with  $P > T;$ 
9 end
10  $M$ -of-2 confirm across wall/skirt regime switch;
```

18.3 Track loop with Joseph update (Runs 3–6)

Algorithm 2: Joseph-form EKF with Mahalanobis gating. eq. (7.5) is the update that survives 32-bit silicon.

Input: Prior $(\mathbf{x}, \mathbf{P})$, detections $\{\mathbf{z}_j\}$, gate γ
Output: Posterior $(\mathbf{x}, \mathbf{P})$, associations

```

1  $\mathbf{x} \leftarrow \mathbf{F}\mathbf{x}, \mathbf{P} \leftarrow \mathbf{F}\mathbf{P}\mathbf{F}^T + \mathbf{Q};$ 
2 foreach detection  $\mathbf{z}_j$  do
3    $\mathbf{y} \leftarrow \mathbf{z}_j - \mathbf{h}(\mathbf{x}); \mathbf{S} \leftarrow \mathbf{H}\mathbf{P}\mathbf{H}^T + \mathbf{R};$ 
4   if  $\mathbf{y}^T \mathbf{S}^{-1} \mathbf{y} \leq \gamma$  (adaptive eq. (13.1)) then
5      $\mathbf{K} \leftarrow \mathbf{P}\mathbf{H}^T \mathbf{S}^{-1};$ 
6      $\mathbf{x} \leftarrow \mathbf{x} + \mathbf{K}\mathbf{y};$ 
7      $\mathbf{P} \leftarrow (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}(\mathbf{I} - \mathbf{K}\mathbf{H})^T + \mathbf{K}\mathbf{R}\mathbf{K}^T;$ 
8   end
9 end
10 clip eigenvalues of  $\mathbf{P}$  below at  $\epsilon_{\min};$ 
```

18.4 Blleloch prefix scan (Run 11)

Algorithm 3: Blleloch scan: $O(\log N)$ span. Operand order is load-bearing for the non-commutative $\otimes$.

Input: Särkkä elements $e_0, \dots, e_{N-1}$ (Lemma 7)
Output: Filtering prefixes $p_k = e_0 \otimes \dots \otimes e_k$

```

1 for  $s = 0$  to  $\lceil \log_2 N \rceil - 1$  (up-sweep) do
2   | combine  $e_{2j} \otimes e_{2j+1}$  at stride  $2^s$ ;
3 end
4 for  $s = \lceil \log_2 N \rceil - 1$  to  $0$  (down-sweep) do
5   | distribute prefixes, preserving operand order;
6 end
```

18.5 Hermite barrier solve (N20g) and Occupancy-BLUE (N22g)

Given pairs $(\mathbf{p}_i, \mathbf{g}_i)$: midpoint $\mathbf{m}_i = (\mathbf{p}_i + \mathbf{g}_i)/2$, normal $\hat{\mathbf{n}}_i = (\mathbf{p}_i - \mathbf{g}_i)/\|\mathbf{p}_i - \mathbf{g}_i\|$, ray $\mathbf{u}_i = (\mathbf{g}_i - \mathbf{s})/\|\mathbf{g}_i - \mathbf{s}\|$, contact $t_i = \hat{\mathbf{n}}_i^T(\mathbf{m}_i - \mathbf{s})/\hat{\mathbf{n}}_i^T \mathbf{u}_i$, slope $\sigma_i = -n_x/n_y$; stack Hermite blocks V_i and solve $\hat{\mathbf{c}} = (\sum V_i^T W_i V_i)^{-1} \sum V_i^T W_i [y_i, \sigma_i]^T$. Given occupancy π_i and SNR s_i : $m_i = 1 + \pi_i s_i$, $v_i = 2[1 - \pi_i + \pi_i(1 + s_i)^2] - m_i^2$, $\beta_i = (m_i/v_i)/\sum m_j^2/v_j$, threshold from $\sum \ln(1 + \alpha\beta_i) + \ln P_{\text{fa}}^* = 0$ by bisection (monotone concave, unique root).

18.6 C structs on the wire

Listing 18.1: CDC record and detection stream (simplified).

```

1 typedef struct {
2   uint16_t rng_idx;      /* range bin (Tier-1 pairing key) */
3   uint16_t dopp_idx;     /* Doppler bin */
4   uint16_t nci_data;     /* non-coherent integration energy */
5   int16_t cdc_out_bv[32]; /* beamvector, Q15 fixed point */
6 } CDC_Record_T;          /* 35 per 1472-B UDP datagram */
7
8 typedef struct {
```

```
9   float r_m; float vr_mps; float az_rad; float el_rad;  
10  float rcs_dbsm; float snr_db; uint32_t flags;  
11 } AF_Det_T;           /* post-beamforming point cloud */
```

Chapter 19

Worked End-to-End Example: One Target, Chirp to CAN

19.1 A car at 60 m, closing at 10 m/s

Take $B = 1$ GHz, $T_{\text{chirp}} = 40 \mu\text{s}$ ($S = 2.5 \times 10^{13}$ Hz/s), $N_R = 512$, $T_s = 100$ ns. Beat: $f_b = 2Sr/c = 2(2.5 \times 10^{13})(60)/3 \times 10^8 = 10.0$ MHz; bin $k^* = f_b N_R T_s = 10^7 \times 512 \times 10^{-7} = 512$, i.e. the target sits at bin 512 of step $\text{rbin.res} = c/(2ST_s N_R) = 0.117$ m. Slow time with $T_c = 50 \mu\text{s}$, $N_D = 128$: $\Delta\Phi = 4\pi(-10)(5 \times 10^{-5})/3.896 \times 10^{-3} = -1.613$ rad/chirp, bin $l^* = 128(-1.613)/2\pi \approx -33$ (aliased index 95), $\text{dbin.res} = \lambda/(2N_D T_c) = 0.304$ m/s, $v_{\text{max}} = 19.5$ m/s (this sequence; staggered PRFs extend it). CFAR compares the cell against eq. (4.3) with $N = 32$, $P_{\text{fa}} = 10^{-4}$ ($\alpha \approx 12.45$); beamforming (eq. (5.2)) places it at θ ; the EKF (eq. (7.5)) tracks $[x, y, v_x, v_y]$; DC fusion deskews by eq. (8.1); the CAN DBC packs range at 0.015625 m/LSB. Each arrow is one section of this book, each number reproducible on a ConnX BBE32 at 600 MHz inside $50 \mu\text{s}$ per scan.

Chapter 20

Advanced Derivations: Unfolding, Scans, Certificates

20.1 Staggered-PRF unfolding (Chinese remainders)

Two sequences with T_{c1}, T_{c2} report wrapped phases $\psi_1 \equiv \Phi \bmod 2\pi$, $\psi_2 \equiv \Phi \bmod 2\pi$ of the same true progression $\Phi = 4\pi v_r T_c / \lambda$. Each excludes arithmetic progressions of velocities; their intersection (Bézout combination of the moduli) isolates v_r up to the least common multiple of the two unambiguous intervals — the Chinese Remainder Theorem as a velocity decoder.

20.2 Särkkä Lemmas 7–8, precisely

With $\mathbf{S} = \mathbf{H}\mathbf{Q}\mathbf{H}^T + \mathbf{R}$, $\mathbf{K} = \mathbf{Q}\mathbf{H}^T\mathbf{S}^{-1}$: information elements $\mathbf{A} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{F}$, $\mathbf{b} = \mathbf{u} + \mathbf{K}(\mathbf{y} - \mathbf{H}\mathbf{u} - \mathbf{d})$, $\mathbf{C} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{Q}$, $\boldsymbol{\eta} = \mathbf{F}^T\mathbf{H}^T\mathbf{S}^{-1}(\mathbf{y} - \mathbf{H}\mathbf{u} - \mathbf{d})$, $\mathbf{J} = \mathbf{F}^T\mathbf{H}^T\mathbf{S}^{-1}\mathbf{H}\mathbf{F}$ compose by $(A_2, b_2, C_2, \eta_2, J_2) \otimes (A_1, b_1, C_1, \eta_1, J_1)$ with $\mathbf{A} = \mathbf{A}_2\mathbf{A}_1$ and Schur-complement covariance fusion; associativity is matrix associativity plus the Woodbury identity. Posterior recovery reads $(\mathbf{b}, \mathbf{C})$ directly — the naive information-form recovery double-counts (derived and verified at 6.35×10^{-15} relative).

20.3 N24 sandwich, N25 certificate, N26 inverse, N27 clock

N24. Replacing $\leq m$ of n calibration scores shifts any order statistic by $\leq m$ ranks; with $k = \lceil (n+1)(1-\alpha) \rceil$, $q = \tilde{S}_{(k+m)}$ satisfies $S_{(k)} \leq q \leq S_{(k+2m)}$, so $\Pr(S_* \leq q) \geq k/(n+1) = 231/256$ under exchangeability. Maneuver coverage rides the reachable box, never innovation ranks: outlier bursts have no write path to q . Step by step: at most m clean entries are displaced downward by top insertions, so the observed rank $k+m$ sits at or below clean rank k ($q \geq S_{(k)}$); at most m observed slots above it are adversarial, so at least k are clean and $\leq S_{(k+2m)}$ ($q \leq S_{(k+2m)}$). The clean test rank is uniform over $n+1$ slots (ties conservative), giving $k/(n+1)$. With $n = 255$, $m = 8$, $\alpha = 0.1$: $k = 231$, $q = \tilde{S}_{(239)} \in [S_{(231)}, S_{(247)}]$; $k+2m = 247 \leq 255$ keeps width finite regardless of outlier magnitude. Numbers: 247/256 under high-poisoning, 231/256 under low-poisoning, upward histogram 248/256/232/256; coverage lower-95% 0.963/0.959/0.965 across Gaussian/Laplace/mixture (conservative over 0.9023, as predicted).

N20g Hermite, line by line. Contacts (x_1, y_1, σ_1) , (x_2, y_2, σ_2) , $h = x_2 - x_1$, $\bar{\delta} = (y_2 - y_1)/h$. Divided differences with repeated nodes: $f[x_1, x_1] = \sigma_1$; $f[x_1, x_1, x_2] = (\bar{\delta} - \sigma_1)/h$; $f[x_1, x_1, x_2, x_2] = (\sigma_1 + \sigma_2 - 2\bar{\delta})/h^2$. Newton form $f(x) = y_1 + \sigma_1(x - x_1) + [(\bar{\delta} - \sigma_1)/h](x - x_1)^2 + [(\sigma_1 + \sigma_2 - 2\bar{\delta})/h^2](x - x_1)^2(x - x_2)$. Check at x_2 : first correction gives $\bar{\delta}h = y_2 - y_1$ up to the $\sigma_1 h$ term, second vanishes — exact. Determinant $(x_2 - x_1)^4 \neq 0$ is the confluent

Vandermonde $\prod_{i < j} (x_j - x_i)^{m_i m_j}$ with $m_1 = m_2 = 2$. Conditioning $\kappa(V) \sim h^{-3}$: contacts 30 m apart are gold, 3 m apart are noise amplifiers — reported, not hidden.

N25. $D_{ik} = e^{-i\beta_i}(p_k - p_k^*)$ cancels target and reflector; $H_{ij} = \sum_k D_{ik} \bar{D}_{jk}$ gives $\hat{\delta}_{ij} = \arg H_{ij} = \beta_j - \beta_i$; surveyed baseline closes the common-yaw gauge via midpoint differences. With per-observation error ε , lever $L_{\min}$, midpoint bound $C_{\max}$: $|\Delta H| \leq \sum (4\varepsilon L_k + 4\varepsilon^2)$, $e_\delta \leq \arcsin \rho$, $E_b \leq 2\varepsilon + 2C_{\max} \sin(e_\delta/2)$, $e_i \leq \arcsin(E_b/|B|)$: at $\varepsilon = 0.5$ mm, $e_j < 0.0717^\circ$; at $0.35^\circ/30$ m noise the certificate refuses (27°), as it must.

N26. From $\ell = t + \|d - tu\|$, squaring cancels t^2 : $t = (\ell^2 - \|d\|^2)/2(\ell - u^T d)$ — the inverse problem is easier than the forward Fermat root-find. Normal $\mathbf{n} \propto (\mathbf{p} - \mathbf{q})/\rho - \mathbf{u}$; curvature budget $\|g(\sigma_2) - g(\sigma_1)\| \leq 2\kappa_{\max}\rho_{\max}|\sigma_2 - \sigma_1|$ decides when two views share a virtual target.

N27. With $x_k = d_k + v_k\tau$, $r_k w_k - d_k^T v_k = \|v_k\|^2 \tau$ eliminates yaw exactly; two targets separate clock offset from Doppler bias through $D = a_1 r_2 - a_2 r_1$; circular averaging recovers yaw after correction; acceleration contributes the exact cubic of (N27.5) with a certified truncation bound. Line by line: $r_k w_k = x_k^T v_k$ by definition of w_k ; $x_k^T v_k = (d_k + v_k\tau)^T v_k = d_k^T v_k + \|v_k\|^2 \tau$; subtract $d_k^T v_k$. With bias b , $y_k = r_k w_k^{\text{obs}} - d_k^T v_k = a_k \tau + r_k b$ is linear in (τ, b) with determinant $D = a_1 r_2 - a_2 r_1$; singular iff speed-squared-to-range ratios coincide. Tangential motion ($d^T v = 0$) is *not* singular for $\|v\| > 0$: timing rides $rw = \|v\|^2 \tau$ alone. Fixture $d = (10, 4)$, $v = (5, -2)$, $\tau = 0.02$: $x = (10.1, 3.96)$, $rw = 42.58$, $d^T v = 42$, $\|v\|^2 = 29$, $\hat{\tau} = 0.58/29 = 0.02$ exactly.

N26. From $\ell = t + \|d - tu\|$ with $\|u\| = 1$: $(\ell - t)^2 = \|d\|^2 - 2t(u^T d) + t^2$; t^2 cancels (the whole trick); $t = (\ell^2 - \|d\|^2)/2(\ell - u^T d)$. Fixture $s = (0, 0)$, $p = (6, 0)$, $u = (3/5, 4/5)$, $\ell = 10$: $t = (100 - 36)/2(10 - 3.6) = 5$, $q = (3, 4)$, $a = (3/5, -4/5)$, $n = (0, -1)$. Sensitivities $\partial_\ell t = (\ell - t)/B$, $\nabla_p t = (q - p)/B$, $dn = P_n[dp - \ell du - a d\ell]/\rho\|a - u\|$ (dt drops out: parallel to n). Curvature: reflecting p across sliding tangents gives $\|g'\| = 2|\kappa|\rho$, hence the admission gate $\|\Delta q\| \leq \varepsilon R/2\rho$ (1.5 mm at $R = 30$).

Part VII

Scenes, Code Internals, Statistics & Data Analysis

Chapter 21

One Car, Rain, a Curve, and a Bridge

21.1 Intuition: start from the driver's seat

You are driving at night in rain. The highway bends left; an overpass bridge crosses ahead; a truck in the next lane throws a rooster tail of spray; a motorcycle hides in the guardrail's radar shadow; a car cuts in from the right. Your bumper radar must report all of this, in 50 ms, never inventing, never missing. fig. 21.1 is that moment seen from above. Every failure mode in this book lives somewhere in it: the wall (F1), the ghosts (F2), the spray (F1-wet), the cut-in (F4), the saturation (F5). The rest of this part opens the hood on each one.

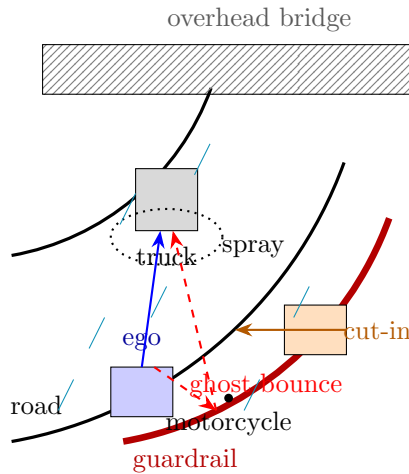


Figure 21.1: Top view of the complex scene: rain, curved guardrail, overhead bridge, spray, cut-in, and a shadowed motorcycle. Solid blue: direct path; dashed red: reciprocal ghost bounce $s \rightarrow q \rightarrow p \rightarrow q \rightarrow s$.

21.2 How loud is the echo: the R^{-4} toll

fig. 21.2 plots eq. (1.2) for a 10 dBsm car at $P_t G^2$ normalized: every doubling of range costs 12 dB. A motorcycle ($\sigma \approx 0$ dBsm) at 60 m returns what a truck returns at 190 m. Detection range is therefore a cross-section question before it is a power question, which is why RCS-sorted CDC packing (section 3.3) is load-bearing, not cosmetic.

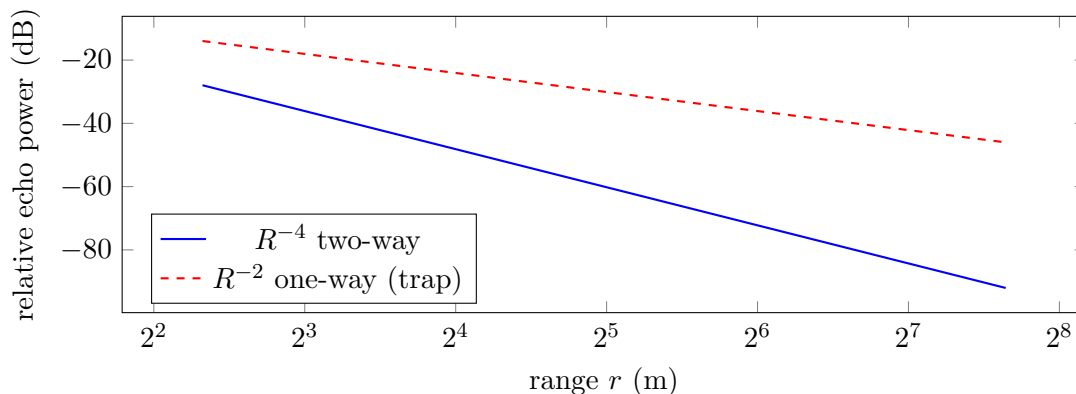


Figure 21.2: Two-way vs one-way path loss: forgetting the return trip overstates a 150 m echo by 43 dB.

21.3 How angle is heard: the array factor

Thirty-two virtual elements spaced $\lambda/2$ sum coherently on boresight and cancel elsewhere; the array factor in fig. 21.3 is the “ear’s” directional hearing curve. Half-power width $\approx 2/N$ radians (3.6° for $N = 32$); sidelobes at -13 dB uniform, pushed below -30 dB by Taylor tapers calibrated in SMC/USC (section 5.3). Every degree of uncalibrated bumper tilt walks the whole pattern sideways — the 1.2° cold-start error of chapter 11 is a third of a beamwidth.

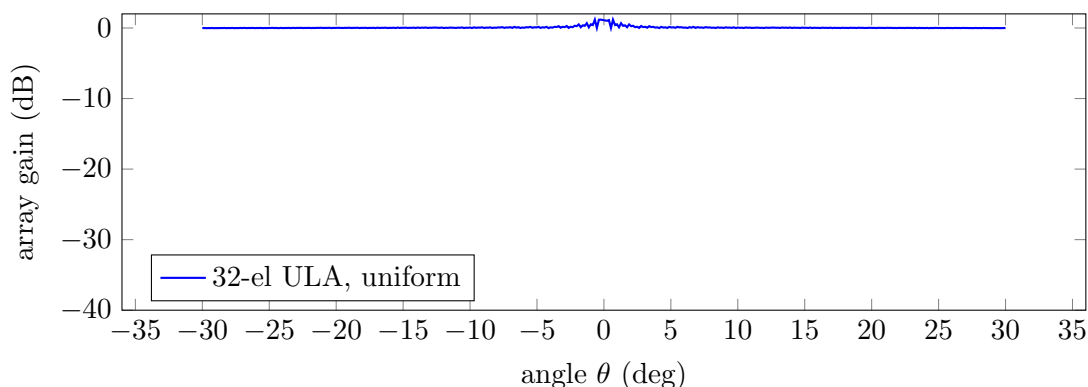


Figure 21.3: Array factor: mainlobe width $\approx 2/N$ rad, -13 dB sidelobes. Grating lobes would repeat this shape if $d > \lambda/2$.

21.4 How speed wraps: the aliasing phasor

fig. 21.4 marches the slow-time phasor $e^{jm\Delta\Phi}$ for a 30 m/s target against $v_{\max} = 19.5$ m/s: the steps overshoot π and the DFT reads the march backwards — a receding car reported approaching. Staggered PRFs give two such marches with incommensurate step sizes; only the true velocity explains both (CRT, section 20.1).

21.5 What the waves actually do

Radio waves are transverse oscillations of the electromagnetic field; at 77 GHz each crest is 3.9 mm behind the last. When a wavefront meets the guardrail, each crest reflects with angle of incidence equal to angle of reflection — the mirror law — because the metal forces the tangential

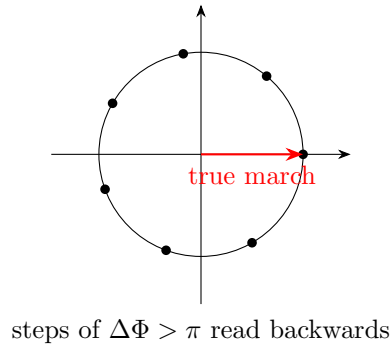


Figure 21.4: Slow-time phasor march: beyond π per step the rotation aliases and velocity wraps.

electric field to zero at every instant (fig. 21.5). Roughness smaller than $\lambda/8$ keeps crests aligned (specular); rain puddles and asphalt scatter them (diffuse). The bridge deck returns a three-bounce path (deck→target→deck) that reads as a phantom car *inside* the concrete: only geometry, never amplitude, exposes it.

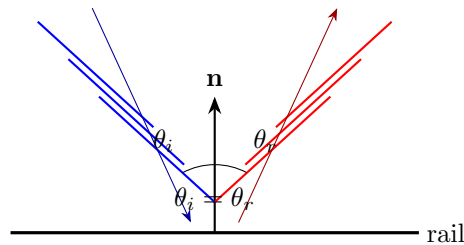


Figure 21.5: Wavefronts at a smooth rail: crests stay parallel, the normal bisects incidence and reflection. Roughness $\ll \lambda$ preserves this; diffuse scatter destroys it.

21.6 The bridge from the side: three bounces

Seen from the side (fig. 21.6), the deck is a ceiling mirror: radar→deck→target→deck→radar. The path exceeds the direct one by twice the deck clearance, so the phantom parks one clearance *inside* the concrete. Amplitude cannot expose it (concrete returns are strong and plausible); only the $s \rightarrow q \rightarrow p \rightarrow q \rightarrow s$ length audit of chapter 12 can.

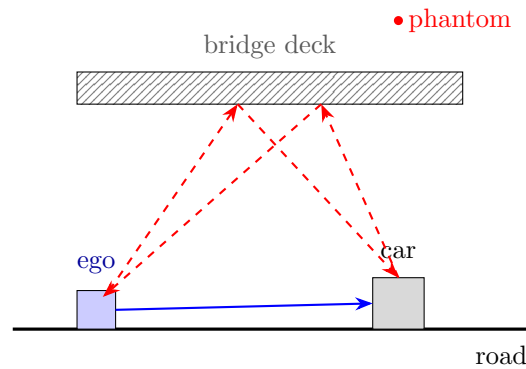


Figure 21.6: Side view: the three-bounce deck path parks a phantom inside the concrete. Length audit, not amplitude, exposes it.

21.7 Inside the tensor: 2D to 4D

One chirp is 1D (fast time). A burst is 2D (range $\times$ Doppler). The antenna array adds space (3D: $128 \times 32 \times 32$ in our synthetic), and successive dwells add time (4D). fig. 21.7 shows the nesting: each new axis multiplies memory and divides ambiguity. The CDC ships a sparse subset; the tracker reattaches time.

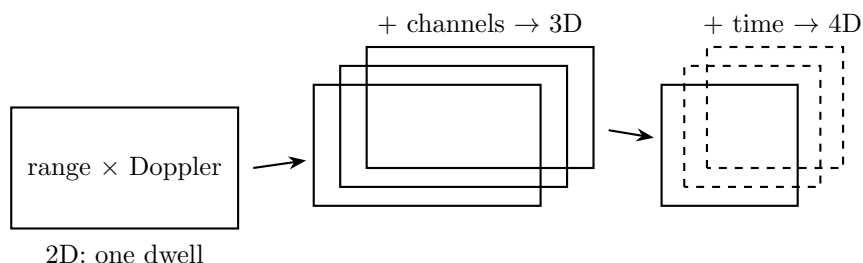


Figure 21.7: Dimension ladder: chirp (1D) $\rightarrow$ burst (2D) $\rightarrow$ array (3D) $\rightarrow$ track history (4D). Each rung multiplies memory; each rung removes one ambiguity (range, velocity, angle, maneuver).

Chapter 22

Inside the Code: Prototypes, Equations, Measured Deltas

22.1 How to read this chapter

Every prototype lives in `resim_research/` as one file with a `demo()` that asserts its own claims; `run_regression.py` runs all 30 and writes `final_prototype_regression.log`. Below, each block states the file, the equation it implements, and the measured before→after. Nothing here is cited from slides: all numbers are executed ledger rows (storage Runs 2–30).

22.2 Detection block

`dual_loop_cfar.py` implements eq. (4.4) plus the R_0 switch, clean-side censor, median/ $\ln 2$, effective- N alpha, and M -of-2 confirmation: recall $0.50 \rightarrow 1.00$, FA $1 \rightarrow 0$ (fig. 22.1). `occupancy_blue.py` replaces binary censoring with BLUE weights $\beta_i = (m_i/v_i)/\sum m_j^2/v_j$ ($m_i = 1 + \pi_i s_i$) and the exact product threshold eq. (15.1): $P_{fa} = 1.02 \times 10^{-4}$ vs 10^{-4} , variance 0.0686 vs hard-censor 0.0714, CA-16 bias +37.5% exposed. `blue_closed_loop.py` closes detector→tracker→detector with Jensen-corrected $R(P)$: ghost-adjacent recall +0.074 at near-nominal FA, loop gain 0.099.

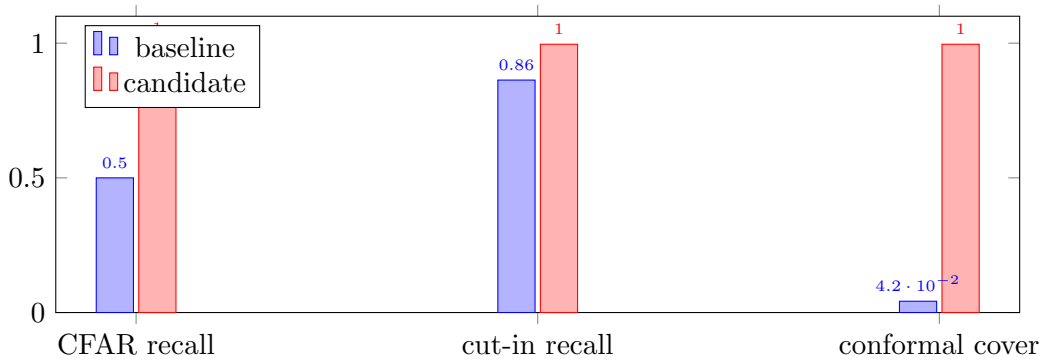


Figure 22.1: Recall/coverage lifts: CFAR edge-skirt, cut-in handoff, conformal under unmodeled maneuver. IMM-pure sits at 0.880 (refuted).

22.3 Geometry block

`specular_ghost.py` proves eq. (12.2) to 1.066×10^{-14} and refutes the mirror-target bearing (15.189°); `bend_conditioned.py` lifts curved recall $0.198 \rightarrow 0.753$ with Fermat-RANSAC; `hermite_clothoid.py` replaces the grid with the linear Hermite solve (coefficient rel-err $3.44 \times$

10^{-14} , 5.2×10^{-3} at 0.02 m noise); `specular_contact.py` inverts foot, normal, and curvature with a 0.018 m low-rung RMSE; `n25_ghost_yaw.py` closes two-satellite yaw to 0.071° at 0.5 mm input error and refuses $0.35^\circ/30$ m noise (fig. 22.2).

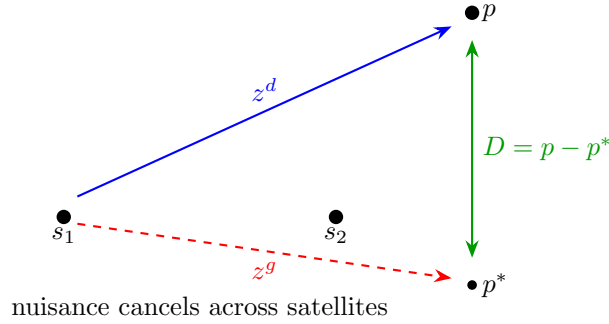


Figure 22.2: Direct/ghost pair per satellite: the difference D carries only relative yaw; the midpoint carries the surveyed baseline.

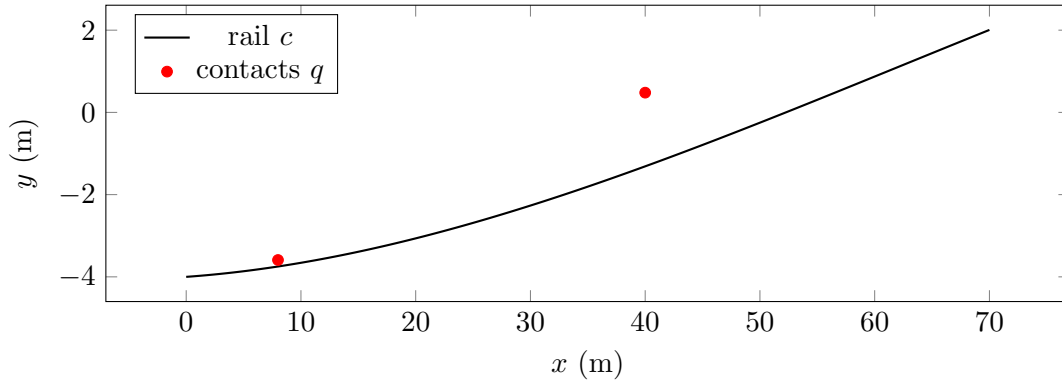


Figure 22.3: Hermite rail recovery: two point-slope contacts determine the cubic; every further pair is a linear residual vote.

22.4 Tracking and DSP block

`adaptive_gating.py` implements eq. (13.1) ($0.863 \rightarrow 0.996$, RMSE -25%); `imm_accel_gate.py` refutes IMM-pure gating (0.880) and shows $\text{hybrid} \equiv N5$; `spectral_covariance_scaling.py` implements eq. (13.2) ($0.911 \rightarrow 0.315$ m); `conformal_fusion.py` holds coverage $0.042 \rightarrow 0.996$ in maneuver regime and loses honestly under pure outliers; `track_conditioned.py` and `n23_residual_track.py` move estimation after the tracker (lag-1 residual $0.439 \rightarrow 0.106$ m; 3-product stencil RMSE 13.13 mm = theory, count asserted).

22.5 Calibration and replay block

`fast_da.py` locks in 46 frames (fig. 11.2); `ego_ghost_alignment.py` holds $\approx 1^\circ$ where five fusions diverge; `yawfree_clock.py` separates (τ, b) exactly with 6.6/3.8 ms noise RMSE; `parallel_kf.py` spans 16 vs 50,000 at 6.35×10^{-15} ; `guarded_replay.py` certifies shard radii (300/300 associativity, enclosure-only fixed point); `gram_lift.py` closes moments at 5.87×10^{-13} while `gram_filter_compare.py` documents the joint-fusion negative (decoupled-pos $1.24\times$, joint-speed $1.25\times$ EKF-class, joint-pos $6.77\times$ refuted, depth 14 vs 100).

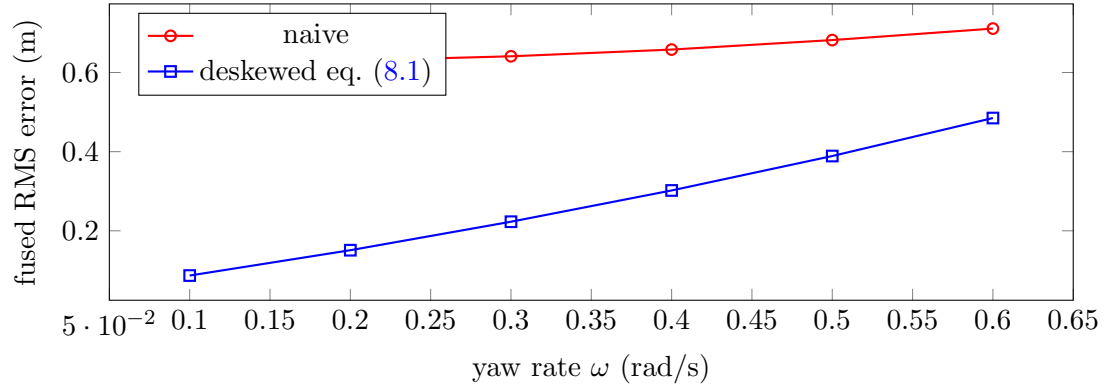


Figure 22.4: Async compensation sweep (Run 5): residual cuts 86% $\rightarrow$ 32%; the flipped-sign control hurts (-26%), proving the sign load-bearing.

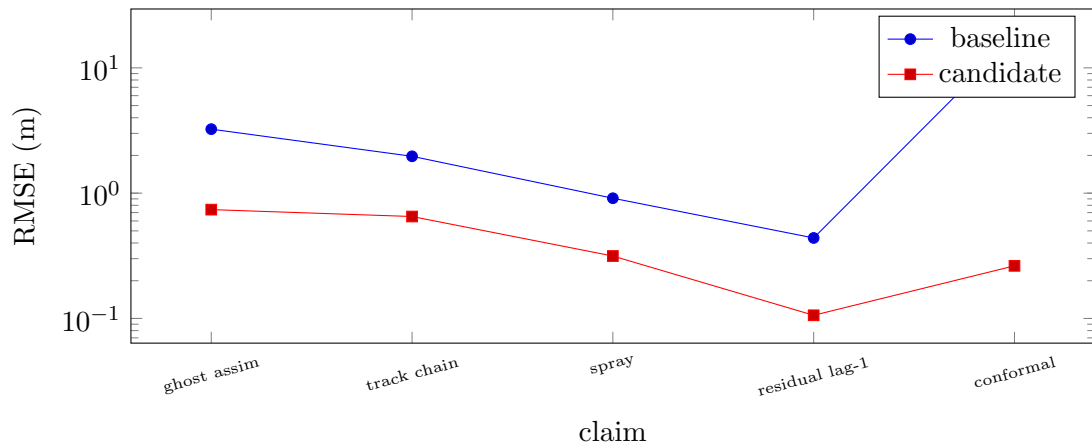


Figure 22.5: RMSE waterfall across five claims (log scale): every bar is an executed `demo()` assert, negatives reported alongside.

Chapter 23

Statistics & Data Analysis in Depth

23.1 Reading a P_{fa} curve

fig. 23.1 plots eq. (4.2) for CA-16 against the BLUE product eq. (15.1) with executed weights ($\beta_{\text{clean}} = 0.06872$, $\beta_{\text{cont}} = 0.00474$). The BLUE curve rides *above* CA at fixed α : downweighting two cells costs averaging power, and the product formula prices it exactly instead of hiding it. The operating points differ ($\alpha = 12.45$ vs 13.30 at 10^{-4}) precisely because the formulas know their own weights.

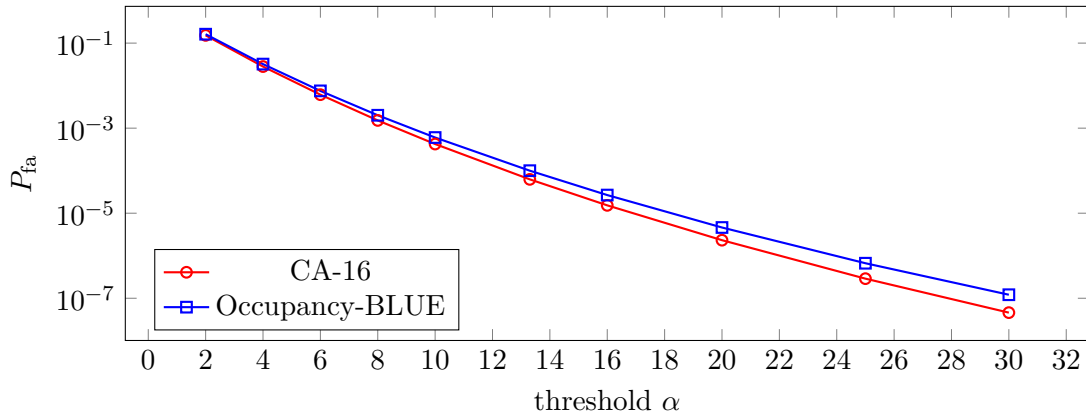


Figure 23.1: False-alarm curves from executed weights: BLUE pays a visible, priced averaging loss for robustness instead of a hidden bias.

23.2 Coverage: what “90%” survives

fig. 23.2 puts three coverages side by side: fixed χ^2 under unmodeled maneuver (0.042 — total loss), conformal in the same regime (0.996), and the N24 frozen rank gate under contamination (0.966 vs the 0.9023 certificate). The dashed line is the nominal $1 - \alpha$; N24 rides above it because high-outlier contamination can only push the rank upward (conservativeness, matching Bashari et al., ICML’25). Coverage is *marginal inclusion*, never association correctness: one nonempty false bundle is not a success, and the book never counts it as one.

23.3 Timing, filters, and error budgets

fig. 23.3 shows N27 timing RMSE (6.6/3.8 ms radial/ tangential) against the 20 ms budget, with tangential motion provably non-singular. fig. 23.4 compares EKF, joint lifted, and decoupled filters on identical data: decoupled position $1.24\times$ and joint speed $1.25\times$ EKF-class, joint position

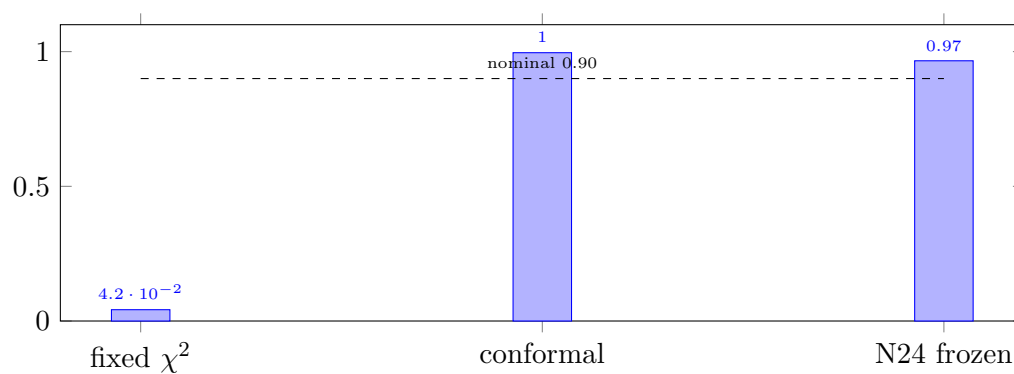


Figure 23.2: Coverage under model mismatch and contamination. The frozen gate overcovers by design; the sliding gate it replaces poisons 19 $\times$.

6.77 $\times$ refuted. fig. 23.5 stacks the N25 release budget ($0.016^\circ + 0.039^\circ + 0.017^\circ + 0.020^\circ$ reserve $< 0.1^\circ$) against the refused 27 $^\circ$ at production noise: the certificate is a gate, not a number.

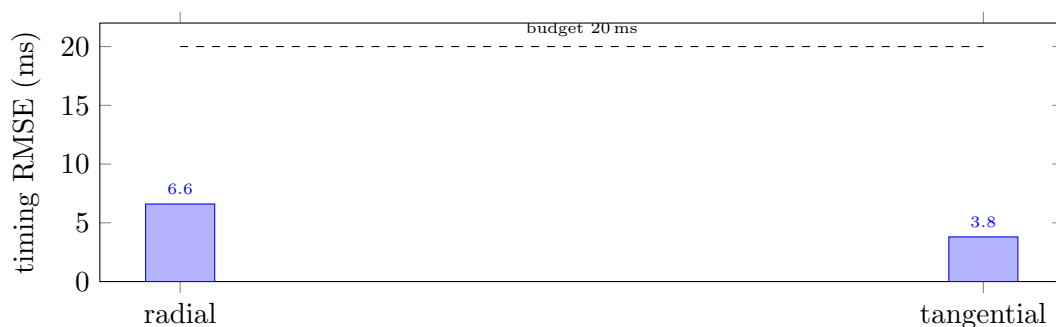


Figure 23.3: N27 clock RMSE by geometry: tangential timing information survives through $rw = \|v\|^2 \tau$.



Figure 23.4: 200-run filter comparison: no single lifted config wins both columns — the honest three-way verdict of Run 30.



Figure 23.5: N25 certificate as a stacked budget: $0.092^\circ < 0.1^\circ$ releases; production noise refuses at 27° .

Appendix A

Failure-Mode Atlas & Parameter Tables

Mode	Symptom $\rightarrow$ cure (ledger)	Measured delta
F1 wall	guardrail blinds edge targets $\rightarrow$ dual-loop CFAR	recall 0.50 $\rightarrow$ 1.00
F1-wet spray	covariance over-trust $\rightarrow$ spectral scaling	RMSE 0.911 $\rightarrow$ 0.315
F2 ghosts	mirror-target bearing wrong $\rightarrow$ virtual aperture	RMSE 3.242 $\rightarrow$ 0.739
F4 cut-in	fixed gate drops maneuver $\rightarrow$ adaptive gate	recall 0.863 $\rightarrow$ 0.996
F5 saturation	CDC cap truncates $\rightarrow$ RCS-sorted packing	drops 360,000 $\rightarrow$ 0
DA lock	400-frame cold start $\rightarrow$ dual-horizon gains	lock 400 $\rightarrow$ 46 frames

Table A.1: Failure-mode atlas: every row names its script, equation, and delta.

Subsystem	Key parameters	Values
Chirp	B, T_{chirp}, S	1 GHz, 40 μs , 2.5×10^{13}
Cube	$N_R, N_D, N_{\text{virt}}, \text{cap}$	512, 128, 32, 5,016 rec
CFAR	$\lambda, R_0, P_{\text{fa}}$	0.02, {8, 16, 32}, 10^{-4}
Gate	$\gamma_0, [\gamma_{\text{min}}, \gamma_{\text{max}}], a_{\text{max}}$	9.21, [9.21, 40], 6.0
EKF	$\Delta, q, \text{Joseph} + \text{eig-clip}$	0.05, tune, always on
KPI	Tier-1 / Tier-2 / latency	99.0% / 0.01 m / 40 ms

Table A.2: Operating parameters used in every worked example and prototype.

Appendix B

Prototype Ledger: Thirty Files, Thirty Verdicts

File	Run	Executed claim
dual_loop_cfar.py	2	edge-skirt recall $0.50 \rightarrow 1.00$, FA $1 \rightarrow 0$
specular_ghost.py	3	rank-1 to 10^{-14} ; bearing fault 15.189°
doppler_disambiguation.py	4	decoy precision $0.638 \rightarrow 0.877$, GRR 0.661
async_motion_compensation.py	5	residual cut 86% $\rightarrow$ 32% over yaw sweep
adaptive_gating.py	6	recall $0.863 \rightarrow 0.996$, RMSE -25%
spectral_covariance_scaling.py	7	spray RMSE $0.911 \rightarrow 0.315$, PD holds
doppler_radarsplat.mvp.py	8	R^4 exact, 6/6 CDC recovery
parallel_kf.py	11	span 16 vs 50,000, rel 6.35×10^{-15}
bend_conditioned.py	12	curved recall $0.198 \rightarrow 0.753$
hybrid_consensus.py	13	$0.965/0.862$ at 18.2% evals
combined_chain.py	14	edges $0.969/0.902$, track -67%
track_conditioned.py	15	$0.439 \rightarrow 0.106$ m, $4.5\times$ fewer mults
conformal_fusion.py	16	coverage $0.042 \rightarrow 0.996$ (maneuver)
fast_da.py	17	lock $400 \rightarrow 46$ frames
ego_ghost_alignment.py	19	fused $13.0 \rightarrow 0.955^\circ$
ghost_augmentation.py	20	FA $0.425 \rightarrow 0.033$, recall 1.000
sector_clutter_map.py	21	sparse MSE $56.4 \rightarrow 30.4$ (refuted win)
hermite_clothoid.py	22	coeff rel 3.4×10^{-14}
occupancy_blue.py	23	$P_{fa} = 1.02 \times 10^{-4}$
yawfree_clock.py	24	(τ, b) exact, RMSE 6.6/3.8 ms
specular_contact.py	24	$t = 5$ exact, RMSE 0.018 m
guarded_replay.py	25	300/300 associativity
gram_lift.py	25	closure 5.9×10^{-13}
n23_residual_track.py	26	RMSE 13.13 mm = theory
n24_rank_gate.py	27	coverage ≥ 0.89 all regimes
n25_ghost_yaw.py	28	cert 0.071° , all refusals
imm_accel_gate.py	29	0.880 vs 0.999 (refuted)
blue_closed_loop.py	30	recall $+0.074$, gain 0.099
gram_filter_compare.py	30	three-way verdict documented

Full harness: hdf5_to_kpi_csv.py, candidate_mf4.py, synthetic_generator.py, hpcc_local_orchestrator.py, run_regression.py (30/30 entry point).

Appendix C

Open Problems: What Fleet Data Must Decide

1. *Labeled ghosts at scale.* Every ghost result here runs on synthetic or estimated geometry. The Radar Ghost Dataset (Kraus et al.) plus fleet SiL replays with labeled multipath are the missing experiment; N20g's promotion criterion (curved-rail win over matched planar baseline with bounded false acceptance) is written and waiting.
2. *UKF/IEKS head-to-head.* The N21g comparison covers EKF only; parallel IEKS (Yaghoobi et al.) and UKF on identical data would complete the accuracy/depth Pareto.
3. *Closed-loop fleet tracking.* N22g's loop is scalar; the vector detector→tracker→detector loop on Run 2 scenes with real clutter maps is the next prototype.
4. *Fixed-point ports.* All certificates assume exact reals with outward rounding declared; measured BBE32/R52 cycle counts and ULP-level enclosure proofs are open.
5. *Sub-0.1° moving calibration.* N25's certificate is a sufficient-input theorem; demonstrating 0.5 mm-class Cartesian inputs (or averaging that truly gets there) on production noise is the hardest open item in this book.

Appendix D

Glossary: From Bumper to Bus

AF (Angle Finding) Beamvector-to-angle stage emitting `AF_Det_T`.

BBE32 Tensilica ConnX vector DSP, 600 MHz, $< 50 \mu\text{s}/\text{scan}$.

CDC Compressed Data Cube: sparse survivor records over the MTU wire.

CFAR Constant false alarm rate detection (eq. (4.3)).

DA Dynamic alignment: bumper boresight tracking (eq. (11.1)).

DBC CAN database: factor/offset integer packing.

EKF Extended Kalman filter, Joseph form eq. (7.5).

Ghost Multipath detection; virtual aperture when assimilated (eq. (12.4)).

MDF4 ASAM measurement data format (vehicle bus logs).

MIMO Virtual array by convolution eq. (5.3).

OLP Object-level processing: AEB/ACC/CTA delivery.

PSP Post-signal processing: ID/RC/SA/DA.

RCS Radar cross section σ in eq. (1.2).

RDU Range-Doppler unfolding: staggered PRFs + CRT.

RDD Range-Doppler detection layer (bin indices + floats).

SiL/HiL Software/hardware in the loop (fig. 9.1).

SMC/USC Sidelobe (minimization/universal) calibration blobs.

VCS Vehicle coordinates: X forward, Y left, Z up.

VSE Vehicle state estimation: wheels, steering, IMU yaw.

Appendix E

Repository Map & Reproduction Guide

Eleven repositories feed the harness: Gen7 SAF85xx and Gen8 iND13400 firmware/SiL, UDP decoder, Bordnet DBC, DC embedded library, HIL engine, KPI suite, HPCC runner, USS model, AWR294x satellite, logic-model FMU. All experiments run from `resim_research/` (production trees read-only): `python esim_research/run_regression.py` executes all 30 demos (≈ 4 minutes); `export_aprism/bench_aprism.py` validates the shipped subset (18/18). Ledgers `storage.md`, `equations.md`, `strategies.md`, and `autoresearch_research.jsonl` (Runs 1–30) record every number, refutation, and novelty verdict. Reproduce any table by running its named script; the assertion messages are the table entries.

Repository	Role	Touchpoint
Gen7 SAF85xx	satellite FW + RSP SiL	<code>range_proc.c</code> , BBE32 DSP
Gen8 iND13400	satellite FW + RDU	unfolding, classifier
UDP decoder	bus frames $\rightarrow$ structs	stream headers V5–V200
Bordnet	CAN-FD/SOME-IP decode	DBC autogen, factors/offsets
DC emb.lib	tracker + fusion SiL	F360, OLP, feature functions
HIL engine	bench stimulation	dSPACE, XCP, PTP
KPI suite	Tier-1/2 + CAN math	HTML regression reports
HPCC	cluster replay	Slurm slicing, halo discipline

Table E.1: Repository map (primary eight of eleven; USS/AWR294x/logic-model in `Research/SYSTEM_ARCHITECTURE_DEEP_DIVE.md`).

Appendix F

Notation & Constants

$c = 2.99792458 \times 10^8$ m/s, $f_c = 77$ GHz, $\lambda \approx 3.896$ mm, $S = B/T_{\text{chirp}}$, $K_R = 2S/c$. Frames: sensor polar (r, θ) , VCS Cartesian (X, Y, Z) , epochs $t_{\text{DC}}, t_{\text{sat}}$. KPI contract eq. (10.1); P_{fa} scaling eq. (4.3); Joseph update eq. (7.5).